

Preserved categorical identification with impaired hue interpolation in color vision deficiency motivates a cortically derived correction filter

Jinil Kim,^a Albert Minkue Cho,^a Jungwoo Seo,^a Jiook Cha^{a*}

^aSeoul National University, Seoul, South Korea

*Correspondence: Jiook Cha (connectome@snu.ac.kr)

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Abstract: Color vision deficiency (CVD) is typically corrected with generic filters designed from a retina-based model. Such filters change how colors appear but improve color discrimination only in some products and tasks. Filter design omits visual cortex, where the retinal loss and the observer's partial compensation combine. How that combination reshapes color representation in CVD remains underinvestigated. Here we show that CVD distorts the geometry of the cortical color representation rather than weakening its overall signal. We scanned two adults with CVD, one with each red-green subtype, and seven healthy controls. All eight colors remained decodable from cortical activity in both CVD participants, whereas the continuous hue geometry departed from controls in both. We modeled each distortion as a hue rotation along two color-space axes and inverted the fit into a personalized stimulus-space filter. Both CVD participants completed a second session of fMRI and behavioral testing to evaluate the filters. Under the individualized filter, the four hue-discrimination thresholds that had exceeded the control range in the first session (three for deutan, one marginal for protan) moved within that range. This small sample cannot establish superiority over a deployed accessibility filter. The direction of cortical change differed across participants and measures, and neither reached the healthy reference. This study identifies a previously uncharacterized geometric distortion of the cortical color representation in CVD

and introduces, to our knowledge, the first cortically grounded framework for individualized color correction.

1 Introduction

Color vision deficiency (CVD) affects approximately 8% of men and 0.4% of women of European ancestry (Birch, 2012). It arises from a shift in the spectral sensitivity of the L- or M-cone photopigment. The magnitude of that shift varies continuously across affected individuals (Bosten, 2019). X-linked polymorphisms in the cone opsin genes underlie the shift (Deeb, 2005; Neitz & Neitz, 2011). The two pigments lie between 1 and 12 nm apart in anomalous trichromacy, compared with roughly 27 nm in normal trichromacy (Bosten, 2019; Neitz & Neitz, 2011). This narrowed separation attenuates the L–M cone-opponent signal without abolishing it. Red–green CVD comprises two subtypes that differ in the direction of the shift. In protan observers the L-cone sensitivity moves toward the M cone, and in deutan observers the M-cone sensitivity moves toward the L cone (Neitz & Neitz, 2011).

Observers who share a diagnostic label differ substantially in how much their color space is compressed. Anomalous trichromats show reduced red–green discrimination, frequent category confusions, and elevated just-noticeable differences along the L–M chromatic axis. The severity of these signs extends from near-normal to near-dichromatic within one diagnostic subtype (Bosten, 2019; Neitz & Neitz, 2011).

Current CVD correction is calibrated to a population-average retina and applies the same transform to every user. Hardware filters are notch lenses. Each notch attenuates a narrow band of the spectrum where the L and M cones overlap. Removing that band widens the difference between the two cone signals (Álvaro et al., 2022). The EnChroma lens cuts three such notches at wavelengths taken from a normal retina (Gómez-Robledo et al., 2018). The lens changes appearance without changing performance. Colors look more saturated and further apart along the red–green axis, while discrimination thresholds move minimally (Somers et al., 2024). The lens produces no clinically meaningful improvement on plate, arrangement, discrimination, naming or sorting tasks, and does not change diagnostic classification (Álvaro et al., 2022; Gómez-Robledo et al., 2018). Two commercial lenses shifted discrimination thresholds in 51 individuals with red–green CVD. For both lenses, variation between observers exceeded the mean effect (Patterson et al., 2022). Software filters start from the same population-average retina. They forward-project a stimulus onto a deficient retina, either by the Brettel–Viénot–Mollon construction for dichromacy (Brettel et al., 1997) or by the parameterized Machado model, which extends the same logic to anomalous trichromacy (Machado et al., 2009). Daltonization inverts these models. It recolors an image, moving red–green differences onto the blue–yellow and lightness axes (Fidaner et al., 2005;

55 Simon-Liedtke & Farup, 2016).

56 Individualized correction has been implemented at the retinal level. It tunes a cone-shift parameter
 57 to a user's responses on plate-style or appearance-matching tests. That parameter can be estimated
 58 accurately, from opsin genotype or from cone spectral sensitivities measured in observers of known
 59 genotype (Deeb, 2005; Stockman & Sharpe, 2000). The retinal parameter predicts color discrimination
 60 imperfectly. Some anomalous trichromats with very small spectral separations discriminate better than
 61 their separation predicts (Neitz & Neitz, 2011). Across observers, the correlation between cone spectral
 62 separation and color discrimination is weaker than expected (Bosten, 2019). A retinal parameter fixes
 63 the input to the visual system. It does not describe the cortical representation that is built from that
 64 input and determines color perception. Filters tuned to an observer's own cone sensitivities trade one set
 65 of confusions for another rather than moving color vision toward normal (Gómez-Robledo et al., 2018).
 66 The remaining step is a filter derived from the user's own cortical color representation.

67 Cortical activity patterns specify both which color was seen and where that color lies among the others.
 68 Cortical color processing is distributed and hierarchical (Conway et al., 2018; Gegenfurtner, 2003; Shapley
 69 & Hawken, 2011). Early visual cortex carries cone-opponent signals. In V1, perceptual hues form spatially
 70 clustered response patterns, while the cardinal directions do not (Parkes et al., 2009). V1 responses are
 71 also selective for hues that fall between the cone-opponent axes (Kuriki et al., 2015). In hV4, a hue
 72 withheld from training can be reconstructed from the responses to its neighbors (Brouwer & Heeger,
 73 2009). The relative positions of colors in these response patterns constitute a geometry, and a color can
 74 be identified in one observer from the response patterns of others (Bannert & Bartels, 2025).

75 What has been measured in CVD is the strength of the color signal. Anomalous trichromats perceive
 76 more red–green contrast than a cone-sensitivity account predicts (Boehm et al., 2014). Deutan ob-
 77 servers adapt to chromatic contrast more than the same account predicts (Basim et al., 2025). Both
 78 results are consistent with a post-receptoral gain that magnifies the weakened L–M signal. Long-term
 79 adaptation to the altered input can produce such an adjustment (Webster, 2015). In V1, color re-
 80 sponses are weaker in anomalous trichromats than in normal trichromats. In ventral V2 and V3 they are
 81 indistinguishable (Tregillus et al., 2021).

82 Color deficiency also changes where colors lie relative to one another. Anomalous trichromats still describe
 83 stimuli with all four hue primaries, but each primary lands on a different stimulus than it does for normal
 84 trichromats (Emery et al., 2021). When dichromats judge the difference between color samples, red
 85 and green fall close together, and the map recovered from those judgments has one axis where normal
 86 trichromats have two (Saysani et al., 2018). When protan and deutan observers judge the difference
 87 between Munsell chips, the map keeps two axes but is dented at yellow and purple-blue. The size of that

dent varies more than tenfold across observers, and some show no dent at all (Ohkoba et al., 2022). All of these are reports. The same relative positions have yet to be measured in the cortical response.

We take the individual's own cortical color representation as the reference and invert it into a correction on the stimulus. Of these regions, hV4 is where the arrangement of responses follows perceptual color space (Brouwer & Heeger, 2009). Our design adopts the premise that shifting an individual's hV4 color geometry toward that of the healthy controls moves their color perception with it.

Three gaps remain. (*Gap 1*) The cortical color geometry of an individual with CVD remains to be described. Existing descriptions come from healthy observers (Brouwer & Heeger, 2009, 2013; Kuriki et al., 2015). (*Gap 2*) The distances between colors have not been measured in the cortical response in CVD. fMRI in CVD has reported response magnitude, including contrast-response gain in anomalous trichromacy (Tregillus et al., 2021) and activation in one dichromat and two observers with achromatopsia (Rina, 2024). (*Gap 3*) A filter inverted from an individual's own cortical representation remains underinvestigated. Existing designs invert models of the retina (Brettel et al., 1997; Machado et al., 2009) or maps of an individual's color judgments (Ohkoba et al., 2022).

We ask whether an individual's own cortical color representation can be turned into a correction for that person. We characterize two adults with CVD, one deutan and one protan, against a reference group of healthy controls (see Methods).

A stimulus-space filter is warranted and tractable in one specific regime, defined by two conditions. The displayed colors must remain decodable from the cortical response. This preserved decodability gives the filter a signal to act on. The continuous arrangement of hues must be distorted, so that there is something to correct. Our questions follow this logic.

1. **Describe.** Does an individual's cortical hue geometry in CVD carry structured distortion, with the individual hues still decodable and their continuous arrangement impaired?
2. **Summarize.** Can a few interpretable parameters capture that distortion, treated as a compact description rather than as physiological measurements?
3. **Correct.** Can a realizable, individualized filter be derived from those parameters?
4. **Validate.** Does the filter move the cortical representation closer to the healthy-control reference than a deployed accessibility filter does?

We expect the continuous arrangement of hues to be selectively impaired while the individual hues remain

decodable. A few interpretable parameters should capture this distortion, and inverting them should yield a realizable, individualized correction.

Together, these steps yield a filter built for one individual. Its input is the cortical signal, its parameters are the fitted description, and its output is a stimulus shift. Each is defined at the level of the single observer, the level at which any correction must ultimately act. This study tests the feasibility of the inversion, namely whether an individual's cortical color geometry can be turned into a realizable filter. It does not test whether per-person correction outperforms a subtype average, which would require several individuals within one subtype.

2 Methods

The analysis proceeded in three stages, summarized in Figure 1C. First, we characterized each CVD participant's cortical color representation with three neural analyses. Leave-one-run-out (LORO) decoding tested whether color classification remained intact (§2.8). Leave-one-color-out (LOCO) decoding tested whether continuous hue interpolation was impaired and yielded a per-hue vulnerability profile (§2.8). Procrustes disparity quantified how far each CVD hue geometry departed from the healthy-control reference (§2.9). Second, we fitted two classes of CVD distortion model by minimizing a combined neural and psychophysical loss. Parameters were selected by a pre-specified three-gate procedure (§2.10–§2.12). Finally, we inverted each fitted distortion numerically to obtain a per-subject stimulus-space correction filter (§2.14). Figure 3 summarizes the fitting and inversion procedure. Each filter was evaluated in a second session that combined fMRI with the two psychophysical tasks described in §2.3.

2.1 Participants

Ten volunteers provided written informed consent and were enrolled. An additional three respondents were excluded during pre-consent screening because they did not meet MRI safety criteria. The study was approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023) and conducted in accordance with the Declaration of Helsinki. All participants reported normal or corrected-to-normal visual acuity. Color vision was screened with the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants who identified 9 or more plates correctly were classified as healthy controls (HC, $n = 7$, 3 female, aged 20–27 years, mean 22.7 ± 2.5). Participants scoring below this threshold were classified as CVD. The classification plates of the same test distinguished the protan from the deutan subtype and indicated anomalous trichromacy rather than dichromacy in both. Two CVD participants

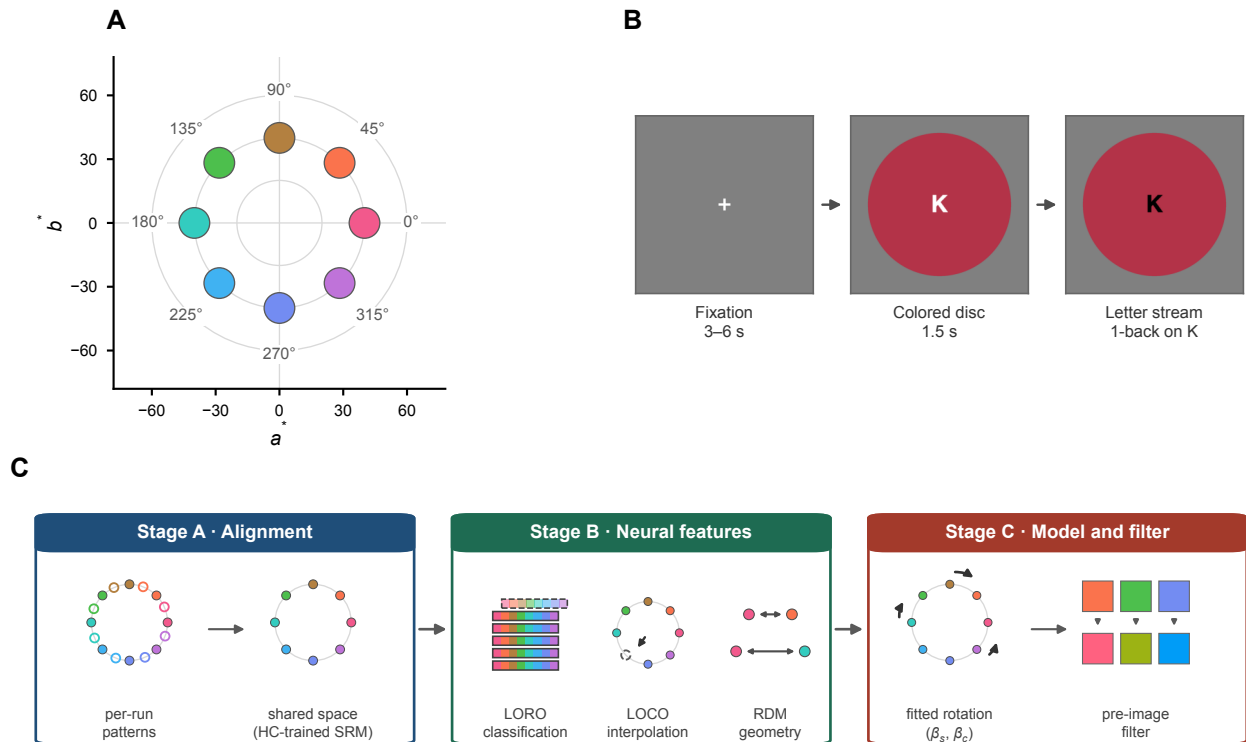


Figure 1: **Stimuli, task, and analysis pipeline.** **(A)** Stimulus set: 8 CIE $L^*a^*b^*$ hues at matched lightness, 45° angular spacing ($L^* = 75$, chroma = 40; red-magenta), shown in the a^*b^* plane. Swatch colors throughout the figure reproduce the displayed stimuli. **(B)** Task: rapid serial visual presentation (RSVP). A fixation period (randomized 3–6 s) is followed by a 1.5 s uniform colored disc; 5–9 trials per color per run (Neurodesign-optimized allocation, identical across participants), six runs. **(C)** Analysis pipeline. Stage A (Alignment): two-stage GLM per-color amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Stage B (Neural feature): LORO classification, LOCO interpolation, and the SRM-space representational dissimilarity matrix (RDM). Stage C (Modeling & filter): a 2-component cortical model with S-cone-axis (β_s) and confusion-axis (β_c) components; its pre-image yields the per-subject stimulus-space filter. In the pipeline glyphs, open circles show per-run patterns before alignment, the dashed strip marks the run held out under LORO, the dashed circle marks the hue held out under LOCO, double-headed arrows depict pairwise representational distances entering the RDM, curved arrows indicate the fitted hue rotation, and the downward arrows map each displayed hue to its filtered counterpart. HC: $n = 7$ (sub-01–07).

completed both sessions, both male. The deutan participant was 24 years old and identified 5 of 14 plates correctly. The protan participant was 25 years old and identified 7 of 14 plates correctly. One further enrolled CVD participant completed the first session but became unavailable before the filter-evaluation session and was excluded from all analyses, leaving nine participants in the analyses. All CVD results are reported as single-case comparisons against the HC distribution with the Crawford and Howell (1998) modified t -test. No population-level inference is made.

2.2 fMRI stimuli and task

Eight chromatic stimuli were defined in CIE $L^*a^*b^*$ space at equal angular spacing in the a^*b^* plane, from 0° to 315° in 45° steps (Commission Internationale de L'Eclairage, 1986). All eight shared a lightness of $L^* = 75$ and a chroma of 40, and they spanned red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler at the same lightness with $a^* = b^* = 0$ served as a null condition. Isoluminance was not determined for individual participants. Stimuli were presented on a BOLDscreen 32 UHD monitor (Cambridge Research Systems) at 3840×2160 resolution, using the Apple XDR Display color preset. The display was used at its factory calibration without further colorimetric verification in the scanner environment.

Stimuli were presented in a rapid serial visual presentation (RSVP) design to sustain attention and fixation. Each trial displayed one uniform chromatic disc, 500 pixels in diameter, for 1.5 s in a pseudo-random order optimized with Neurodesign (Durnez et al., 2018). Two letters were presented sequentially at the center of the disc, and participants pressed a key on an MRI-compatible button box whenever the letter K appeared twice consecutively. Inter-stimulus intervals were randomly drawn from $\{3, 4.5, 6\}$ s. Each run comprised 72 events and lasted approximately 7 min, and each session comprised six runs. The experiment was implemented in PsychoPy 2022.2.5 (Peirce et al., 2019).

The schedule was optimized for detection power rather than for equal counts, so color counts varied within a run. Each chromatic stimulus occurred 5–9 times per run and 38–47 times across the six runs, and the remaining 14–20 events per run presented the gray filler. Every participant received the same optimized schedule, so this allocation contributes a common component to all individual estimates. The general linear model estimated one amplitude per color and run regardless of how many trials contributed, so all downstream analyses operated on a balanced $6 \text{ (run)} \times 8 \text{ (color)}$ design.

2.3 Psychophysical tasks

All participants completed two psychophysical tasks on the day of their first scanning session. The first task measured just-noticeable differences (JNDs) in hue discrimination, and the second was an eight-alternative forced-choice (8AFC) color-identification task. Both CVD participants repeated both tasks under each filter in the second session (§2.15). Both tasks ran outside the scanner under normal room lighting, on a built-in Liquid Retina XDR display set to the same Apple XDR Display color preset.

Just-noticeable difference (JND). On each trial two chromatic discs appeared simultaneously to the left and right of fixation, and participants judged whether the two were the same or different. Each disc had a radius of 96 pixels in a 1024×768 window, their centers were separated by 260 pixels, and their positions were randomized across trials. A fixation cross preceded each trial for 0.5 s, and the discs remained on screen until the participant responded. The two hues were separated by a fraction t of the pair's full separation, interpolated in CIE LCh , and t served as the staircase level. Two staircases per pair were interleaved from starting levels $t = 0.8$ and $t = 0.5$, both following a 1-up/1-down rule and converging around the level yielding 50% "different" responses (Levitt, 1971). The step size was 0.15 for the first two reversals and 0.03 thereafter. A staircase terminated at eight reversals and at least 20 trials, extended by two further reversals whenever the last six reversal levels had an SD above 0.10. The threshold was defined as the mean of the last six reversal levels obtained after the first two. Because the two discs always differed, no same trials were included, and the threshold may therefore reflect response bias as well as perceptual sensitivity.

Eight hue pairs were selected to span the S-cone axis and each subtype's confusion axis. Yellow–purple lies along the S-cone axis, and blue–purple and red–orange are positioned on either side of it. Orange–yellow and yellow–green bracket the deutan confusion axis, and green–blue and cyan–magenta bracket the protan confusion axis. Red–cyan, separated by 180° , served as a control against ceiling effects. Pair selection was fixed on these grounds before any threshold was collected. For each pair we computed the JND ratio $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$, in which $\bar{\gamma}_{\text{HC}}$ denotes the HC mean threshold. Ratios above 1 indicate elevated hue-discrimination difficulty relative to HC. Because all participants were measured on the same display, the ratio is unaffected by any difference between that display and the one used in the scanner. The fitting loss of §2.11 uses the raw thresholds rather than this ratio.

8AFC color identification. Each trial began with a 1.0 s fixation, followed by a uniform chromatic disc of 500 pixels in diameter for 1.5 s and a 0.5 s blank interval. Participants then used the arrow keys to navigate a two-by-four grid of the eight scanner stimulus colors and selected the hue that had been

presented. Each color was shown as a disc 100 pixels in radius, with 240-pixel horizontal and vertical spacing. The positions of the eight options were reshuffled on every trial. Trials timed out after 10s without a response. No feedback was provided. Each run comprised 64 trials, eight per color, in random order. Identification accuracy served as an independent psychophysical readout, held out from the fitting loss.

2.4 MRI acquisition and preprocessing

Imaging was performed on a Siemens 3T MAGNETOM Cima.X scanner with a 64-channel head coil. A whole-brain T1-weighted MPRAGE volume served as the anatomical reference. It was acquired sagittally at 1 mm isotropic resolution, with a repetition time of 2000 ms, an echo time of 2.36 ms, a field of view of 256 mm, a bandwidth of 220 Hz per pixel, and GRAPPA acceleration factor 2. Functional coverage was restricted to posterior occipital cortex with 24 interleaved oblique slices aligned perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009), spanning 48 mm. The functional sequence used a repetition time of 1.5 s, an echo time of 30 ms, a flip angle of 70°, and 2 mm isotropic voxels over a 192 mm field of view on a 96×80 matrix. Acceleration was GRAPPA factor 2 without multiband, the bandwidth was 1447 Hz per pixel, and phase encoding ran right to left. Each run yielded 288–292 volumes. A gradient-echo field map with echo times of 3.88 and 6.34 ms was acquired in each session and associated with the functional runs in the BIDS metadata.

Data from the first session were converted to BIDS format with the ezBIDS web service (Levitas et al., 2024), accessed in October 2025, which also defaced the anatomical image. The second session was converted with dcm2bids 3.2.0 (Boré et al., 2024) in June 2026, and its anatomical image was defaced with Quickshear (Schimke & Hale, 2011), the same defacing algorithm employed by ezBIDS. Both routes used dcm2niix for DICOM conversion. Preprocessing followed a custom pipeline rather than a standard container, and Supplementary §S1 records the routes that were evaluated and rejected before it was adopted. All steps used FSL 6.0.5.1 and FreeSurfer 7.2.0. The T1w image was skull-stripped with bet2 (Smith, 2002), and functional images were registered to it by mutual-information coregistration initialized from the scanner header, using `mri_coreg` (Maes et al., 1997; Wells et al., 1996). Normalization to the ICBM 152 Nonlinear Asymmetric 2009c template (MNI152NLin2009cAsym) used a twelve-parameter affine transform (FLIRT) followed by nonlinear warping (FNIRT) (Andersson et al., 2007; Jenkinson et al., 2002), yielding 2 mm isotropic BOLD data in MNI space. Each functional volume was resampled once by this composed transform, without slice-timing correction, head-motion correction, susceptibility distortion correction, or spatial smoothing. Because a single transform served every volume in a run, the same spatial interpolation applied throughout. Head-motion correction composes a volume-specific

rigid term with that transform. The resampling then attenuates high spatial frequencies and leaves a residual signal change that follows the estimated displacement, raising the error variance of each voxel time series (Grooten et al., 2000). Multivoxel pattern analysis operates on those per-voxel values, so every neural endpoint was computed under both pipelines and both are reported. The second pipeline estimated each volume's rigid term with MCFLIRT (Jenkinson et al., 2002) and composed it with the normalization transform before the single resampling. The head-motion quality-control record appears in Supplementary §S1.

2.5 ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the probabilistic atlas of visual topography of Wang et al. (2015), resampled to 2 mm and thresholded at a probability above 50%. V1, V2, and V3 each merge the ventral and dorsal subdivisions of the atlas. The threshold was applied to each ROI independently, so a voxel assigned to more than one ROI was retained in each. Each ROI was then intersected with that participant's BOLD brain mask, and the resulting coverage is reported in Supplementary §S2. Mean voxel counts across the nine analyzed participants were 665 for V1, 458 for V2, 105 for V3, and 64 for hV4, with standard deviations of 209, 113, 19, and 18.

Response amplitudes were estimated with a two-stage general linear model (Brouwer & Heeger, 2009, 2013; Dale, 1999). In the first stage a finite impulse response model with eight delays, spanning 0 to 10.5 s, was fitted to the concatenated runs without drift terms. Averaging the fitted responses over voxels and over the eight colors gave one hemodynamic response function per ROI and participant.

In the second stage that function and its temporal derivative were convolved with per-color event sequences and fitted to each run separately by ordinary least squares, alongside a linear and a constant drift term for that run. The result was one amplitude for every voxel, color, and run, arranged as a $6 \times 8 \times V$ array per participant.

Amplitudes were aligned across runs by a within-subject Procrustes rotation (Gower, 1975). For each of runs 2 to 6, the 8×8 orthogonal matrix Q minimizing $\|X_1 - X_r Q\|_F$ was applied to that run's color-by-voxel matrix X_r , where X_1 is run 1. Scaling was not permitted, so the rotation acts within the eight-dimensional space of color responses and leaves each voxel's identity unchanged.

2.6 Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings $W_i^{\text{sr}} \in \mathbb{R}^{V \times k}$ (V voxels, k latent dimensions) and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i^{\text{sr}}, S} \sum_{i=1}^N \|X_i - W_i^{\text{sr}} S\|_F^2 \quad \text{s.t.} \quad W_i^{\text{sr}\top} W_i^{\text{sr}} = I_k \quad (1)$$

The orthonormality constraint makes the mapping from the shared space back into each participant's voxel space an isometry. Distances and angles in the shared space are therefore those of the reconstructed color patterns. The latent dimension k was selected per ROI by leave-one-subject-out cross-validation over the candidate values 2 to 6 (Supplementary §S4). SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}}^{\text{sr}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. The reduced dimension was chosen by seven-fold leave-one-subject-out cross-validation, withholding one HC participant from SRM training on each fold. Candidate values from 2 to 6 were ranked on each fold by three quality metrics, namely RDM split-half reliability, cross-subject RDM correlation, and reconstruction error, and the value with the best mean rank was selected. This procedure yielded $k = 4$ for V1 and V2 and $k = 3$ for V3 and hV4.

2.7 Forward encoding model

Both classification (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus as a $F = 6$ -dimensional channel vector $\mathbf{c}(\theta) = [c_1(\theta), \dots, c_F(\theta)]^\top$, where $c_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$ is a half-wave rectified squared cosine tuned to preferred hue θ_k , with preferred hues spaced equally at 60° . Stacking the eight stimulus channel vectors row-wise gives the channel design matrix $C \in \mathbb{R}^{8 \times F}$. Decoding and encoding optimize different targets and therefore use different estimators of the channel-to-voxel weight matrix $W \in \mathbb{R}^{F \times V}$.

Decoding (stimulus hue from voxels). Decoding recovers the stimulus hue from voxel responses. We estimated W by ordinary least squares, taking the pseudoinverse of the channel design matrix C . For a held-out voxel pattern $\mathbf{x}$, channel activations were reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$. The decoded hue corresponded to the hue angle, among all 360 integer angles, whose basis vector correlated most

strongly with $\hat{c}$ (Brouwer and Heeger, 2009; Figure 2). For eight-way classification, the decoded angle was assigned to the nearest of the eight stimulus hues. We compared this decoding approach with five alternative decoders, as reported in Supplementary §S7.

Encoding (voxel responses from channels). Encoding predicts the voxel responses from the channels as $\hat{X} = CW$, where $\hat{X} \in \mathbb{R}^{8 \times V}$. We estimated W by ridge regression. A single penalty α was chosen for each fit by generalized cross-validation (Golub et al., 1979), minimizing the GCV score averaged over the voxels of the ROI (Supplementary §S5). Regularization is warranted because the few stimuli and the correlated basis channels would otherwise overfit W and degrade prediction of held-out responses (Kay et al., 2008; Naselaris et al., 2011). This ridge prior is isotropic and imposes no spatial structure on the voxels.

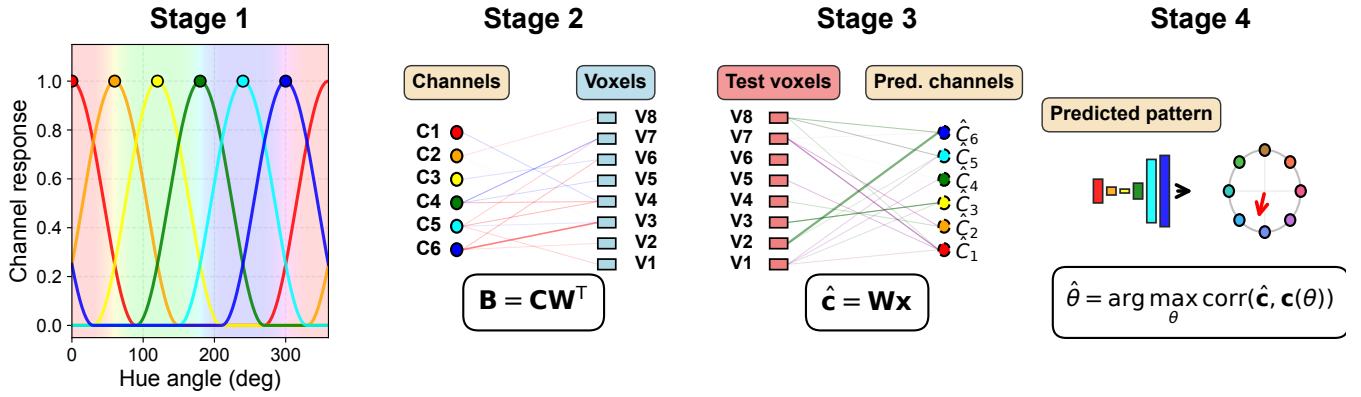


Figure 2: **Forward encoding model schematic.** Stage 1: each displayed hue is encoded as channel activations by six half-wave rectified squared-cosine basis functions, with preferred hues spaced at 60 degrees (Brouwer & Heeger 2009). Stage 2 (training): the channel-to-voxel weights W are fit by ordinary least squares for hue decoding, or by GCV-regularized ridge regression for voxel-response prediction (encoding). Stage 3 (testing): for a held-out voxel pattern, channel activations are reconstructed as $\hat{c} = Wx$. Stage 4: the decoded hue is recovered as the angle, among all 360 integer hues, whose basis vector has the highest Pearson correlation with the reconstructed channel pattern. The same model serves both decoding schemes (Section 2.8): cross-run classification (LORO) and cross-color interpolation (LOCO). Their basis and correlation-based selection are identical, and only the held-out test set differs. Decoding uses the OLS pseudoinverse weights.

2.8 Two decoding schemes: cross-run classification and cross-color interpolation

Two complementary cross-validation schemes applied the forward encoding model to characterize each participant’s hue processing. Leave-one-run-out (LORO) removed one run at a time. Every hue therefore

remained in the training set, and the scheme tested the categorical representation of the eight hues, measured as color classification. Leave-one-color-out (LOCO) removed all six runs of one hue at a time. The held-out hue therefore never appeared in training, and the scheme tested the continuous representation, measured as hue interpolation. Both schemes were computed on the Procrustes-aligned amplitudes of §2.5. The fold construction, the leakage control on the run-level alignment, the cross-subject scheme, and the decoding and voxel-prediction metrics are given in Supplementary §S6.

Cross-run classification (LORO) Color classification was assessed by leave-one-run-out cross-validation. For each of the six runs, W was estimated on the remaining five runs from mean-subtracted amplitudes, and the eight color patterns of the held-out run were decoded by the correlation readout of §2.7. Primary performance was the eight-way classification accuracy averaged across the six folds, against a chance level of 0.125. Single-case comparisons against the healthy controls used the Crawford and Howell (1998) modified t -test. The test is two-tailed here because the hypothesis is preservation, whereas the interpolation tests below are one-tailed.

To test whether hue representations were sufficiently shared across participants, cross-subject transfer of the classifier was evaluated separately. All eight hues enter training here, so this analysis extends LORO across participants and has no interpolation counterpart. The encoder W maps the six channels onto the voxels of the participant it was fitted to. Applying it to a second participant therefore requires the two voxel spaces to correspond, which the SRM-aligned space supplies. Transfer to a held-out control used a leave-one-control-out encoder trained on the remaining six controls, giving 28 participant-by-ROI cells. Transfer to CVD used an encoder trained on all seven controls, giving 8 cells. The two destinations were compared by a Mann–Whitney U test with rank-biserial correlation as the effect size.

Cross-color interpolation (LOCO) Continuous hue interpolation was assessed by leave-one-color-out cross-validation. Each of the eight hues was held out in turn. W was estimated from the remaining seven hues across all six runs, pooled into a 42-sample design matrix, and the held-out hue was predicted by the same correlation readout.

The primary ROI for interpolation was hV4. Brouwer and Heeger (2009) reconstructed novel colors from V4 and VO1, the ROIs where their interpolation was strongest. LOCO was computed in all four ROIs. Crediting a ROI with interpolation required the healthy-control mean to exceed the color-label permutation null defined below. This criterion tests specificity to color identity rather than overall decodability and identifies the ROIs in which a CVD reduction can be meaningfully interpreted. Single-case CVD comparisons are reported for ROIs meeting the criterion. Values for all four ROIs are given in

the Results.

Adjacent accuracy The primary cross-color decoding metric was adjacent accuracy, the proportion of predictions falling within 45° of the target hue. This tolerance corresponds to one hue step. Values were averaged across the six runs of each test color and range from 0 to 1. The readout selects among all 360 integer hues, so a prediction drawn uniformly from that output space falls within tolerance on 91 of 360 draws, giving a chance level of 0.25. Exact accuracy, which requires the prediction to fall in the correct one of eight 45° bins, has a chance level of 0.125. Healthy-control performance was tested against an empirical null of 1,000 random color-label permutations rather than against the analytic chance level. Single-case HC–CVD comparisons at hV4 used the Crawford and Howell (1998) modified t -test, one-tailed and lower.

Vulnerability profile The eight-dimensional vector of per-hue adjacent-accuracy values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant’s *hue vulnerability profile*. Per-hue group comparisons (Figure 4C) used the Crawford and Howell (1998) modified t -test, one-tailed and uncorrected. Effect sizes are reported as the case-control index $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}}$, which equals $t\sqrt{(n+1)/n}$ (Supplementary §S17).

2.9 Representational geometry

Procrustes disparity measures how far a participant’s whole representational geometry lies from the healthy-control reference in SRM-aligned space. Disparity was the Procrustes distance between a participant’s SRM-aligned $8 \times k$ pattern and that reference. Both patterns were centered on their column means and scaled to unit Frobenius norm. Disparity was then the residual Frobenius norm after the optimal orthogonal rotation.

The HC disparity distribution was estimated by a leave-one-HC-out procedure. Each HC participant was compared to the mean of the remaining six, which keeps that participant out of its own reference. Individual CVD disparity was compared to this distribution by the Crawford and Howell (1998) modified t -test, one-tailed upper at $\alpha = 0.05$.

Two estimates of the distribution were computed (Supplementary §S9). In the first, the shared SRM space is trained on all seven HC, so each HC reference is in-sample while the projected CVD participant is out-of-sample. In the second, the space is retrained on six HC at each fold, and the held-out HC is projected by SVD on the same terms as each CVD participant. Each CVD participant receives the mean of its seven fold disparities, whereas each HC participant contributes the single value from the fold

that held it out. The control distribution therefore retains fold-to-fold variance that the CVD statistic averages out, which makes the comparison conservative.

The two estimates serve different purposes. Every other stage of the pipeline operates in the all-HC space, so that space defines the representation the filter is fitted in. Testing whether a participant departs from the healthy-control distribution requires the held-out participant to be projected on the same terms as the CVD participants, which the symmetric estimate provides. We report both and treat the symmetric estimate as the inferential test (table S6). Disparity was additionally recomputed with two distances that use no shared response model, which establishes that it tracks a property of the data rather than of the alignment procedure (Supplementary §S11). Because both patterns are scaled to unit norm before the rotation, disparity is insensitive to any difference in overall response magnitude, which is examined separately (Supplementary §S3).

2.10 Cortical distortion model

The distortion model parameterizes the angular displacement $\delta\theta(\theta)$ that maps the displayed hue angle θ to the angle perceived by a CVD observer. A retinal-family comparison model, retinal-plus-gain (R+C), was evaluated on the same criterion and is defined in Supplementary §S13.

The second model parameterizes the distortion as a sum of two cosine components, one on the S-cone axis and one on each subtype's confusion axis:

$$\delta\theta(\theta; \beta_s, \beta_c) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}}) \quad (2)$$

where θ_{conf} is the hue angle assigned to each subtype's confusion direction, 16° for protan and 150° for deutan. Both angles are fixed before fitting. They are defined in a cone-opponent convention and applied to the nominal $L^*a^*b^*$ hue angle of the stimuli, which is an approximation we adopt for the whole model. β_s parameterizes distortion along the S-cone axis ($\theta = 90^\circ$) and β_c along the individual confusion axis. Each component has a fixed direction and a free amplitude, so the model spans distortions in the plane defined by those two axes. Both signs of β_c are unconstrained. We constrain $\beta_s \geq 0$ as a weak directional prior that fixes only the sign, not the magnitude, of the S-cone-axis term, consistent with the perceptual hue circle of protan and deutan observers being indented at yellow and purple-blue, the two ends of the S-cone axis Ohkoba et al., 2022. Model adequacy was assessed by the procedure in §2.12; results are reported in §3.4.

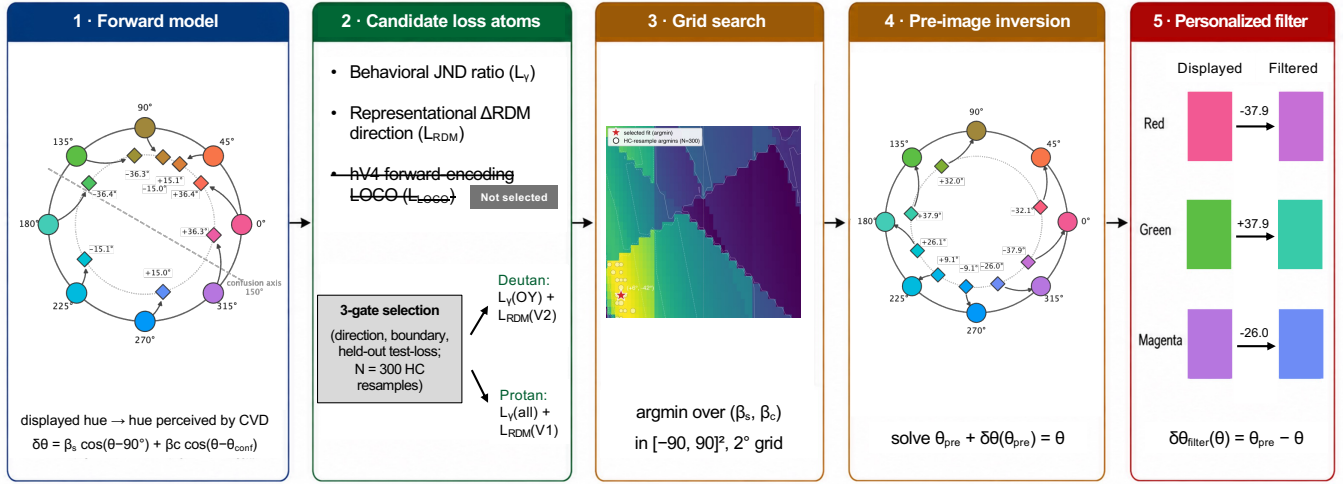


Figure 3: Fitting a per-subject cortical distortion and inverting it into a stimulus-space filter. **(1)** A two-parameter forward model expresses the per-hue distortion as $\delta\theta(\theta) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}})$: an S-cone-axis component (β_s) plus a family-specific confusion-axis rotation (β_c ; $\theta_{\text{conf}} = 16^\circ$ protan, 150° deutan). Outer ring, displayed hue; inner ring, hue perceived under the fitted distortion; labels give the signed displacement. **(2)** Three candidate loss atoms quantify CVD–HC differences. A three-gate procedure (direction, boundary saturation, held-out test-loss over $N = 300$ HC resamples; §2.12) selected a different combination for each participant (deutan, $\gamma_{\text{OY}} + L_{\text{RDM}}^{(\text{V2})}$; protan, $\gamma_{\text{all}} + L_{\text{RDM}}^{(\text{V1})}$); the LOCO atom entered neither selected combination. **(3)** A grid search over $(\beta_s, \beta_c) \in [-90^\circ, 90^\circ]^2$ at 2° resolution minimizes the z -scored composite to the selected fit $\hat{\beta}$. The surface shown is the deutan participant’s; both landscapes and their resample uncertainty are shown in Supplementary §S14. **(4)** The fitted forward map is numerically inverted: for each displayed hue θ , the corrected stimulus θ_{pre} solves $\theta_{\text{pre}} + \delta\theta(\theta_{\text{pre}}) = \theta$. **(5)** The individualized filter is $\delta\theta_{\text{filter}}(\theta) = \theta_{\text{pre}} - \theta$; three of the eight hues are shown for the deutan participant, and the full eight-hue filter for both participants is in Figure 6. Hue wheels and swatches are computed from each participant’s fitted parameters; the schematic reports no new statistics.

2.11 Inverse fitting

Per-subject distortion parameters (β_s, β_c) were estimated by finding the distortion that, when applied to HC-derived neural representations, best reproduces the observed CVD responses. Three loss families served as the fitting objective. A specific instance of a family — for example, the aggregate JND ratio or the Δ RDM loss at a single ROI — is a *loss atom*; candidate objectives are combinations of atoms, selected as described in §2.12.

Psychophysical JND ratio (L_γ). A distortion changes the perceived angular separation of a hue pair, and a pair that is compressed becomes harder to discriminate. We modeled the predicted threshold for a pair as $\hat{\gamma} = \bar{\gamma}_{\text{HC}}(d_{\text{phys}}/d_{\text{perc}})$, where d_{phys} is the separation of the two hues on the display, d_{perc} is their

separation after the candidate distortion of Eq. 2, and $\bar{\gamma}_{\text{HC}}$ is the HC mean threshold for that pair. The loss is the squared difference between the predicted and the observed CVD threshold, standardized by the HC standard deviation s_{HC} for that pair and summed over the pairs entering the atom:

$$L_{\gamma} = \sum_p \left(\frac{\hat{\gamma}_p - \gamma_{\text{CVD},p}}{s_{\text{HC},p}} \right)^2 \quad (3)$$

The atoms available to the fitting were single-pair terms and one aggregate term. The single-pair candidates were the pairs that bracket each participant's confusion axis and the S-cone pair (§2.3), namely orange–yellow, yellow–green, and yellow–purple for the deutan participant and green–blue for the protan participant. The aggregate term γ_{all} sums over all eight tested pairs. Which atom entered each participant's objective was decided by the procedure in §2.12.

Per-pair RDM difference (ΔRDM). The optimal rotation in Procrustes disparity (§2.9) removes the information about which colors are displaced. Inverting the distortion requires that information. We therefore expressed the same comparison as a difference of representational dissimilarity matrices (Kriegeskorte et al., 2008), which retains the displacement of every color pair. For each participant and ROI the RDM was the 8×8 matrix of pairwise correlation distances ($1 - \text{Pearson } r$) among the reduced color patterns of that ROI, and the per-pair difference was $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$, in which $\overline{\text{RDM}}_{\text{HC}}$ is the mean RDM over the seven HC participants.

Representational ΔRDM cosine (L_{RDM}). For a candidate distortion, a simulated difference $\Delta\text{RDM}_{\text{sim}}$ is obtained by applying $\delta\theta$ to the HC mean RDM and comparing it to the observed ΔRDM . Each distorted hue is assigned to the nearest of the eight 45° bins, so $\Delta\text{RDM}_{\text{sim}}$ is built by reindexing the entries of the HC mean RDM. A candidate distortion therefore acts as a permutation of the eight hues, and distortions that map to the same permutation are indistinguishable to this term. The loss is the cosine dissimilarity between the observed and simulated 28-element upper-triangle vectors:

$$L_{\text{RDM}} = 1 - \cos(\Delta\text{RDM}_{\text{sim}}, \Delta\text{RDM}_{\text{obs}}) \quad (4)$$

Because the loss is a cosine, the magnitude of ΔRDM does not enter.

The loss computes ΔRDM in a PCA-reduced space ($K = 6$). Across the 28 pairwise entries this estimate agreed with the SRM-aligned estimate used for disparity at V1, V2 and hV4 ($r = 0.77\text{--}0.89$), and less closely at V3 ($r = 0.39\text{--}0.58$), so the reduction does not determine the pattern the loss fits. The PCA basis is the production basis. The full selection and fitting procedure was additionally repeated with

the RDM atoms computed in the SRM-aligned space, and the agreement between the two reductions is reported with the fits (§3.4, Supplementary §S14).

hV4 LOCO voxel-prediction (L_{LOCO}). A third atom scored how well a candidate distortion accounted for the CVD participant’s own held-out responses at hV4. It was available to every candidate objective and is defined in Supplementary §S14. The selection procedure admitted it in neither participant.

Composite loss. The atoms differ in scale, since L_γ is built from threshold ratios while L_{RDM} and L_{LOCO} are bounded dissimilarities. Each atom was therefore evaluated over the full parameter grid and standardized to zero mean and unit standard deviation across that grid. The composite loss is the sum of the standardized atoms divided by $\sqrt{n_a}$, where n_a is the number of atoms. Standardization sets the relative weight of the atoms and makes any constant rescaling of an individual atom immaterial.

Grid search. Parameters were searched by exhaustive grid at 2° resolution, with β_s over $[0^\circ, 50^\circ]$ (26 values) and β_c over $[-50^\circ, 50^\circ]$ (51 values), giving 1,326 cells. For the R+C comparison model (Supplementary §S13), g was searched over $[0, 3]$ in steps of 0.05 (61 values). Per-subject loss combinations and parameter selection are described in §2.12.

2.12 Parameter selection

Per-subject loss combinations and model class were selected by a three-gate procedure. Two resampling schemes over the seven HC references supplied the inputs, a 5-train/2-test resample repeated $N = 300$ times and a strict 7-fold leave-one-HC-out. Both used the same loss atoms.

Gate 1: Separation precondition. Admission required the CVD loss value to differ from the HC leave-one-out distribution by a Cohen’s d of magnitude 0.5 or greater, in either direction. Atoms below that threshold were excluded from combination.

Gate 2: Boundary saturation. Among admitted combinations, those in which at least half of the resample solutions saturated the grid boundary were rejected, since boundary saturation indicates model misspecification. Combinations were also rejected when the recovered parameters collapsed across resamples, defined as an interquartile range above 50° or a sign reversal between the training and test argmin exceeding 5° .

Gate 3: Held-out test-loss. Surviving combinations were ranked by the median composite test-loss $\bar{L}_{\text{test}}$ across the $N = 300$ resamples. For each resample the composite loss was evaluated at the training argmin on the two held-out HC and z-normalized relative to the training grid, so a lower $\bar{L}_{\text{test}}$ indicates generalization to unseen HC references. Ties were broken by the test-loss interquartile range, which measures the stability of that generalization estimate. These two quantities were the only ranking criteria.

Further quantities were recorded for each candidate outside the ranking. The held-out neural loss $\bar{L}_{\text{RDM}}^{\text{LOO}}$ from the strict 7-fold leave-one-out evaluates a candidate’s neural atoms with the psychophysical term excluded. It therefore isolates what the neural terms alone contribute to the fit. Parameter stability was summarized by the interquartile range of the recovered $(\hat{\beta}_s, \hat{\beta}_c)$, by the fraction of resamples recovering the modal 45° bin, and by the strict 7-fold parameter range. The HC false-positive rate and the distribution of fitted parameter magnitudes over the same HC refits are likewise reported descriptively in Supplementary §S14. To attribute the fit between the two loss families, each selected objective was additionally refit with the psychophysical atom alone, with the neural atom alone, and with the RDM atom removed. These ablation refits entered no selection decision and are reported descriptively in §3.5.

2.13 Identifiability and recovery

Four pre-specified checks were run on each production candidate before any parameter was reported.

Parameter recovery. This test establishes whether a known distortion can be recovered at the present sample size and noise level. Synthetic CVD responses were generated at a known ground truth by passing that distortion through each HC participant’s encoder. Noise matched that participant’s residual structure, using the top 20 principal components of its spatial covariance and an AR(1) correlation of 0.3 across runs. The grid fit was re-run on each synthetic dataset, giving $n = 140$ refits per candidate from seven HC encoders and twenty noise realizations. Recovery was summarized by f_{10° , the fraction of refits within 10° of ground truth on both axes, against a pass criterion of $f_{10^\circ} \geq 0.5$ with $|\text{bias}| < 10^\circ$. The refit repeats the grid search rather than the full selection procedure, so it measures recoverability at a fixed loss combination.

Origin-recovery null. This test records the spread the procedure returns when no distortion is present, which sets its per-axis uncertainty floor. The same synthesis was run with ground truth at $(0^\circ, 0^\circ)$ and each HC participant’s real JND. It returns no p -value and was excluded from the test pool.

HC pseudo-CVD specificity. This test asks whether a HC participant placed in the CVD position yields a comparable estimate. Each HC participant’s real amplitudes were substituted into the CVD slot in turn, giving a seven-member null and a one-sided rank-distance percentile.

Color-label permutation. This test asks whether the estimate depends on which color evoked which response. The eight color labels of the real CVD amplitudes were permuted $N = 1000$ times, with the same permutation applied across ROIs and the HC pool left unchanged, giving a one-sided p_{perm} .

The three test-bearing checks across the two production candidates gave six tests, corrected together under the Benjamini–Hochberg false-discovery rate ($\alpha = 0.05$). Outcomes are summarized in §3.4 and reported in full in Supplementary §S14.

2.14 Stimulus-space filter

The correction filter for each participant is the pre-image of the fitted transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the eight displayed hues θ_k we solved $T(\tilde{\theta}_k) = \theta_k$ for the input angle $\tilde{\theta}_k$, which gives a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. Under the model, a CVD observer viewing $\tilde{\theta}_k$ then obtains the cortical hue representation that a healthy control observer obtains when viewing θ_k .

Each pre-image was located by Brent root-finding on the residual $T(\theta) - \theta_k$, searched within $\pm 60^\circ$ of the target and refined to a bracket tolerance of 10^{-9} degrees. The transform is not guaranteed to be invertible for arbitrary parameters, so every solution was verified by forward-substitution. All eight hues resolved to within 10^{-3} degrees of their target for both deployed parameter sets, and the procedure rejects a parameter set that leaves any hue unresolved.

2.15 Filter evaluation

Both CVD participants completed a second scanning session that compared the individualized filter and a deployed macOS accessibility filter against the unfiltered Session-1 baseline. The individualized filter was the per-subject 2-component stimulus-space pre-image, with parameters frozen from the main analysis and therefore out-of-sample. Acquisition, comparator, run-count, and single-case-inference details are given in Supplementary §S15.

On the second-session data we recomputed, within each condition, the primary Session-1 interpolation metric (LOCO adjacent accuracy, §2.8), LORO classification, SRM Procrustes disparity into the HC

shared space, and the representational dissimilarity matrices. As a secondary encoding index we also computed the forward-tuning correlation, the Pearson r between the predicted and the observed voxel pattern at each held-out hue, averaged over the eight hues. It indexes the encoding fit rather than the decoding accuracy that carries the comparison, and is reported as corroboration only.

Each condition's neural index was reported as a standardized single-case effect size (Cohen's d against the HC distribution) rather than an inferential p , because the grand-mean permutation null is biased for this design. In both CVD participants, the JND and 8AFC tasks (§2.3) were repeated under each filter, each compared to the unfiltered Session-1 baseline and to the HC distribution. This interleaved, cross-session psychophysical comparison is descriptive. Identification accuracy is reported with Wilson score intervals over the $n = 64$ trials of each condition. With two participants all filter comparisons are within-person, and the comparison between the two filters remains open.

2.16 Reproducibility

All analyses were conducted in Python 3.10 with numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0 (Pedregosa et al., 2011), and BrainIAK 0.11. The correction families, the single-case test, and the permutation counts used throughout are collected in Supplementary §S16. Random seeds were fixed for every permutation test and bootstrap. Most analyses used seed 42, while the parameter-recovery synthesis used 31337 and the color-label permutation used 27182, and each script records its own value. Analysis code and preprocessing scripts are openly available at https://github.com/Transconnectome/colorBlind_analysis. The neuroimaging data are held under the access conditions imposed by the approved protocol, which are stated in full in the Data and Code Availability section.

3 Results

3.1 Categorical identification is preserved while hue interpolation falls to chance at hV4 in both CVD cases

Under leave-one-run-out (LORO) cross-validation both CVD participants classified all eight hues above the 0.125 chance level at every ROI (Figure 4A; §2.8). This satisfies the precondition for a corrective stimulus-space filter. The lowest CVD value was 0.375 at hV4, three times the chance level. Classification was likewise preserved under head-motion correction, where the lowest CVD cell across the four ROIs

was 0.229, still 1.8 times chance, and no single-case contrast approached significance in either pipeline (table S2). Repeating the readout in the SRM-aligned space reproduces the pattern (Supplementary §S8). This parallels psychophysical evidence that the perception of large color differences remains largely intact in anomalous trichromacy (Boehm et al., 2014).

An encoder built from the healthy controls also decoded hues from the CVD participants' cortical responses. Accuracy on their held-out runs averaged 0.432 over the eight participant-by-ROI cells, above the 0.125 chance level ($t(7) = 6.51$, $p < 0.001$). The same encoder reached 0.526 over the 28 cells of the held-out controls (Mann–Whitney $U = 163.5$, $p = 0.052$, rank-biserial $r = 0.46$). A channel-to-voxel weight matrix estimated in the controls therefore predicts CVD responses. Single-case comparisons of each participant's own accuracy against the controls appear in Supplementary §S17, where all eight cells lie within the control range.

Hue interpolation, by contrast, succeeded at hV4 alone in the healthy controls (adjacent accuracy 0.456 ± 0.039 , mean $\pm$ SEM over $n = 7$; $p = 0.011$ against 1,000 per-subject color-label permutations). Interpolation was measured by leave-one-color-out (LOCO) decoding, for which the analytic chance level is 0.25 (§2.8). All four ROIs exceeded 0.25, but the permutation null lay near 0.35 (Supplementary §S1), and V1 to V3 stayed within it (0.393, 0.357, and 0.339; $p = 0.164$, 0.424, 0.586). The localization held under head-motion correction, with hV4 again above its null (0.451, $p = 0.023$) and V1 to V3 within theirs (0.283, 0.381, 0.315; $p \geq 0.228$; table S2). This replicates Brouwer and Heeger (2009) and fixes hV4 as the ROI where a CVD reduction in interpolation is interpretable.

Both CVD participants fell below the control distribution at hV4, significantly so in the protan participant (adjacent accuracy 0.13; Crawford & Howell $t = -3.04$, $p = 0.012$, $d_{cc} = -3.25$) and at $d_{cc} = -2.02$ in the deutan participant (0.25; $t = -1.89$, $p = 0.054$), against a control mean of 0.46 (Figure 4B). The magnitude of this reduction was sensitive to preprocessing: under head-motion correction both single-case contrasts returned $p > .10$, and only the primary-pipeline protan contrast excluded zero from its effect-size interval (table S2). The positions relative to the controls nevertheless stayed consistent, with both participants below the control mean in each pipeline, all seven controls above the protan participant, and at least five above the deutan participant. Categorical identification and continuous interpolation thus dissociate within the same hV4 voxels and runs, a pattern that indicates a deficit specific to hue interpolation rather than a general loss of signal quality.

This reduction concentrated on the S-cone intermediate hues. Both participants gave zero adjacent accuracy at hV4 for blue, purple, and magenta, giving identical single-case statistics ($d_{cc} = -2.06$, -0.85 , and -1.57 ; Figure 4C). Per-hue tests reached $p \geq 0.051$ at every hue (Crawford & Howell, one-tailed, uncorrected and exploratory), with blue the largest deviation. The per-hue accuracies form

each participant's hue vulnerability profile $\mathbf{v} \in [0, 1]^8$.

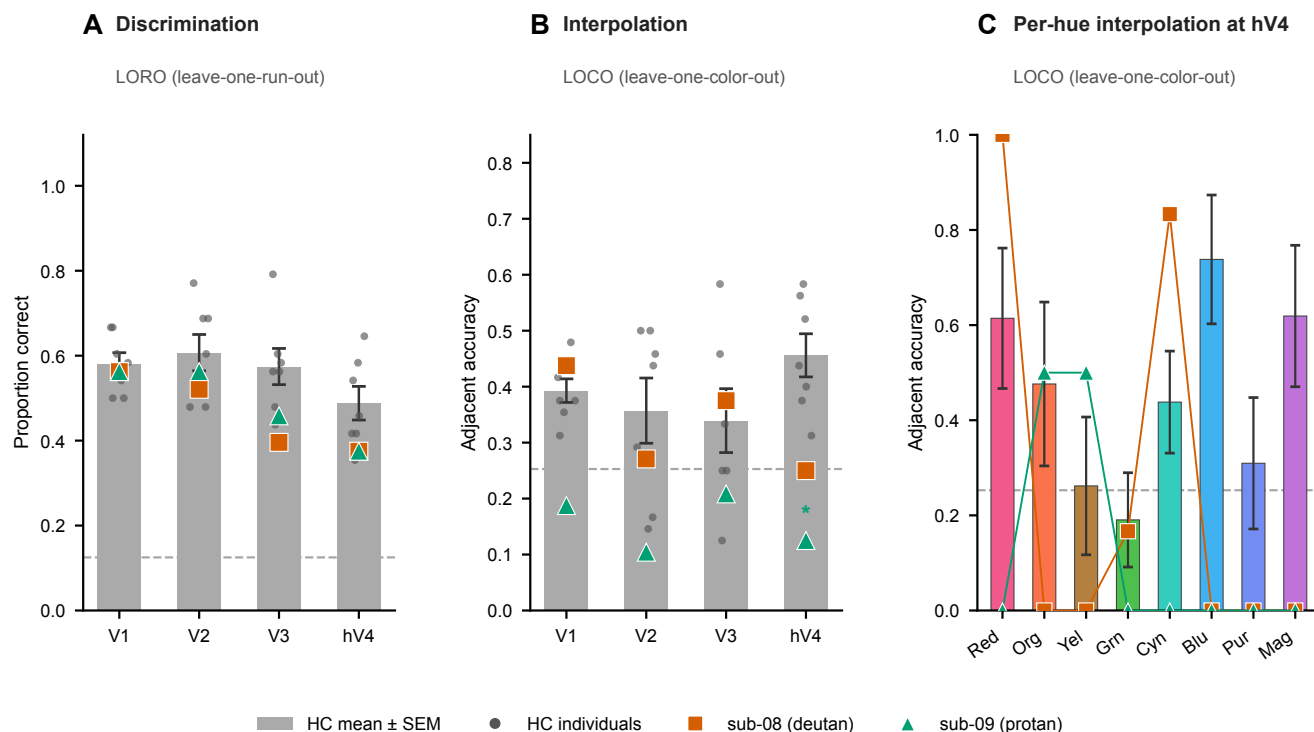


Figure 4: Color classification and hue interpolation across V1–hV4 in the two CVD participants and healthy controls. All panels are computed on the Procrustes-aligned amplitudes with $n = 7$ healthy controls at every ROI. Gray bars give the control mean $\pm$ SEM, the orange square the deutan participant, and the teal triangle the protan participant. **(A)** Eight-way classification accuracy from leave-one-run-out cross-validation of the forward encoding model. The dashed line marks the chance level of $1/8$. Single-case Crawford–Howell comparisons are two-tailed here, since the hypothesis is preservation. **(B)** Adjacent accuracy from leave-one-color-out cross-validation, the proportion of predictions falling within 45° of the target hue. The dashed line marks the analytic chance level of 0.25 . Accuracies are shown for all four ROIs. Asterisks give one-tailed single-case Crawford–Howell comparisons and appear only at ROIs whose control mean exceeds the color-label permutation null. **(C)** Per-hue adjacent accuracy at hV4. Per-hue comparisons are one-tailed, uncorrected, and exploratory across the eight hues.

3.2 Hue geometry departs from the control reference in both CVD cases

Procrustes disparity from the healthy-control reference rose in both CVD participants (Figure 5). The region carrying the largest deviation remained V1 in the protan participant across both pipelines and varied in the deutan participant, at V2 in the primary pipeline and at V1 under head-motion correction. Under the symmetric leave-one-subject-out reference, the protan V1 elevation reached significance in the primary pipeline ($p = .045$) and the remaining cells stayed within the control range. We therefore report

the regional attribution as descriptive and treat the disparity structure as the quantity that the neural loss term evaluates (Supplementary §S1).

Full test statistics and effect sizes for both estimators and both pipelines are reported in table S6 and table S1, and the validity checks for the geometric comparison appear in Supplementary §S10.

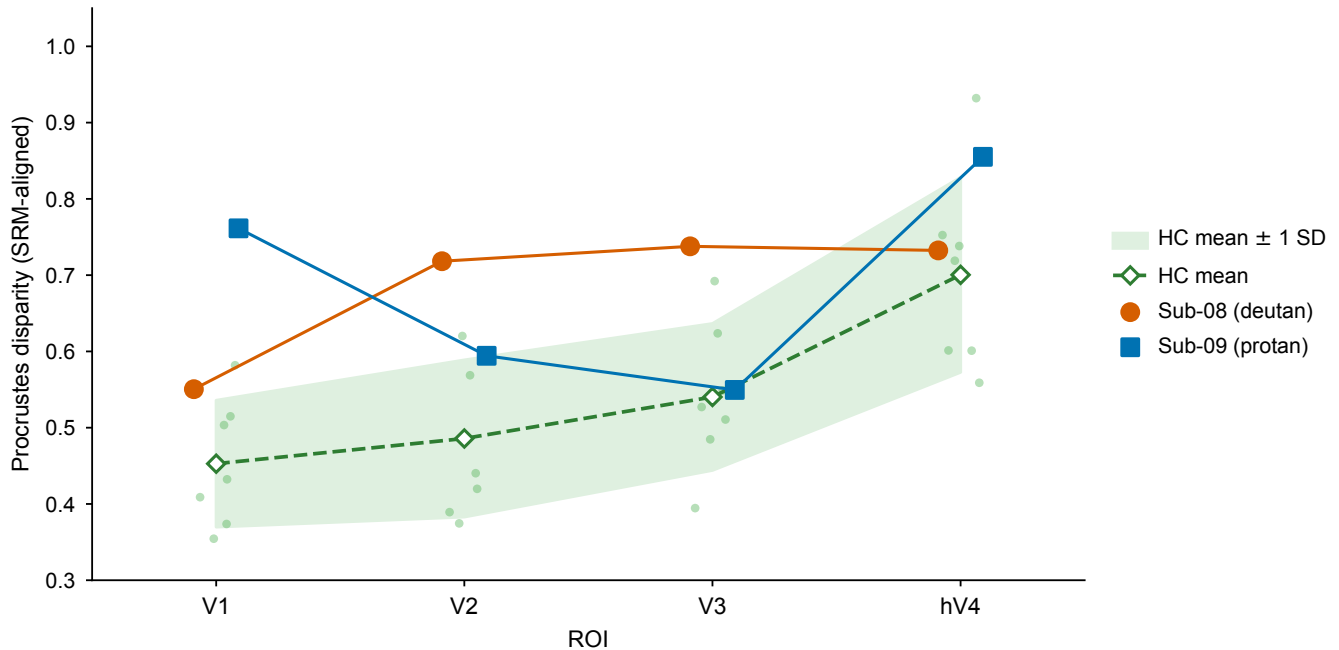


Figure 5: **Procrustes disparity to the healthy-control reference.** Disparity per participant and ROI, indexing the distance of the whole hue geometry from that reference. The gray band gives the HC group mean ± 1 SD ($n = 7$) and the dots give the individual HC leave-one-out values. Single-case comparisons used the Crawford–Howell modified t against the HC leave-one-out distribution, one-tailed. Asterisks are omitted; region-wise tests for both preprocessing pipelines are given in Supplementary §S1. Test statistics under both references are given in table S6.

3.3 Hue-discrimination thresholds are elevated on each participant's own confusion axis

In the just-noticeable-difference (JND) task (§2.3), each CVD participant's threshold elevations aligned with the confusion axis of their own subtype. In the deutan participant the two pairs bracketing the deutan axis were elevated, orange–yellow ($\gamma = 3.02$, $z = +4.15$) and yellow–green ($\gamma = 3.10$, $z = +4.32$), as was the S-cone pair yellow–purple ($\gamma = 2.87$, $z = +6.70$). In the protan participant the elevation was confined to green–blue, which brackets the protan axis ($\gamma = 1.95$, $z = +2.36$). Here γ is the ratio of the participant's threshold to the healthy-control mean and z is the deviation in control standard deviations

($n = 7$).

The remaining pairs lay within the control range in both participants ($|z| \leq 1.23$). This includes red–cyan, the 180° control pair, where both participants discriminated more finely than the controls ($\gamma = 0.45$). Mean $|z|$ over the eight pairs was 2.24 in the deutan and 0.90 in the protan participant. Values for all eight pairs appear in Supplementary §S12.

3.4 A common cortical model fits both CVD cases

The separation precondition determined the candidate grid. The RDM atom entered only at ROIs where it separated the participant from the control distribution (Section 2.12), which left all four ROIs in the deutan participant and V1 alone in the protan participant (separation $d = 2.31, 1.94, 0.86$, and 2.19 across V1 to hV4, against $0.81, -0.23, -0.48$, and -0.24). The psychophysical axis was enumerated in full. Both winning loss combinations paired a JND atom with a Δ RDM atom, and the LOCO family entered neither.

For the deutan participant ($\theta_{\text{conf}} = 150^\circ$), $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ ranked first among the 25 combinations passing the boundary-saturation gate and yielded $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$; the held-out test-loss was $\bar{L}_{\text{test}} = -2.36$ (IQR = 2.15), against -1.14 for the competing β_s -dominant candidate ($38^\circ, -10^\circ$). For the protan participant ($\theta_{\text{conf}} = 16^\circ$), $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$ ranked first among the four combinations passing the same gate and yielded $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$, with a held-out test-loss of -1.54 (IQR = 1.42).

Both fits generalized to held-out control data. On each of the seven leave-one-out folds the CVD representational geometry was predicted more closely by the fitted distortion than by the null distortion $(\beta_s, \beta_c) = (0^\circ, 0^\circ)$, and on held-out loss the selected cell ranked in the top 5% of the 1,326-cell parameter grid for the deutan participant and the top 8% for the protan. Held-out losses for both fits are listed in table S13.

Each participant’s selected fit is the global minimum of its loss surface on the full data. The deutan estimate is robust across the 300 resamples, the seven leave-one-out folds, and both basis reductions, and its confusion-axis magnitude (42°) exceeds the 26° recovery uncertainty of the procedure. The protan estimate, however, is basis-sensitive, returning $(2^\circ, +24^\circ)$ under the PCA basis but $(32^\circ, 0^\circ)$ under the SRM basis, and its 24° magnitude lies within the recovery uncertainty. None of the six pre-specified identifiability tests of parameter recovery, CVD specificity, and color specificity reached significance after Benjamini–Hochberg correction, so the fitted optima are reported as descriptive embeddings of the distortion rather than physiological point estimates (Supplementary §S14).

By contrast, the retinal-family comparison model does not represent this distortion. Its single gain rescales the Machado cone shift along the confusion axis alone, cannot express the S-cone displacement on which the interpolation deficit concentrates, and saturated the gain boundary in both participants ($g = 3.0$ and 2.95). Scored on the same held-out composite, its protan fit reached $\bar{L}_{\text{test}} = -0.86$ against -1.54 for the 2-component model (Supplementary §S13).

3.5 The neural term relocates the protan fit and sharpens the deutan fit

The neural term moved the fitted optimum in the protan participant, sharpened it in the deutan participant, and narrowed the resample spread in both. The confusion-axis component of the protan distortion is carried by the neural data. The RDM atom moves the estimate from the S-cone-dominant ($26^\circ, +4^\circ$) of the psychophysical fit to the confusion-axis-dominant ($2^\circ, +24^\circ$). At that estimate the psychophysical term matches the no-distortion control ($\Delta L = +0.01$, below it on 3 of 7 folds), whereas the RDM term falls 0.47 below it.

In the deutan participant the RDM atom adds precision rather than direction. The psychophysical atom alone already settles at ($16^\circ, -44^\circ$), against ($6^\circ, -42^\circ$) for the combined fit. Adding the RDM atom lowers the boundary-saturation rate from 23% to 9.3%, and a neural-only fit reaches ($4^\circ, -26^\circ$) on the same side of the confusion axis.

The neural term also narrowed the resample spread in both participants. In the deutan participant the HC-resample IQR fell from ($18^\circ, 6^\circ$) to ($8^\circ, 2^\circ$) under the PCA basis and to ($10^\circ, 4^\circ$) under the SRM basis. In the protan participant it fell from ($6^\circ, 4^\circ$) to ($0^\circ, 0^\circ$) under the PCA basis and to ($0^\circ, 2^\circ$) under the SRM basis, where the optimum sits at ($32^\circ, +0^\circ$) rather than at the PCA solution. Therefore, the neural term contributed information that the psychophysical atoms alone did not carry.

3.6 Per-subject stimulus-space filter

Each filter is the exact numerical pre-image of that participant's fitted 2-component transform (Section 2.14). The filter parameters and their psychophysical evaluation are computed from behavioral measures and from frozen fits, and therefore stand outside the comparison between the preprocessing pipelines.

For the deutan participant ($\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -42^\circ$, $\theta_{\text{conf}} = 150^\circ$) the per-hue correction ranges from 9.1° to 37.9° in magnitude, with a mean of 26.3° (Figure 6). For the protan participant ($\hat{\beta}_s = 2^\circ$, $\hat{\beta}_c = +24^\circ$, $\theta_{\text{conf}} = 16^\circ$) it ranges from 6.1° to 24.6° , with a mean of 16.2° .

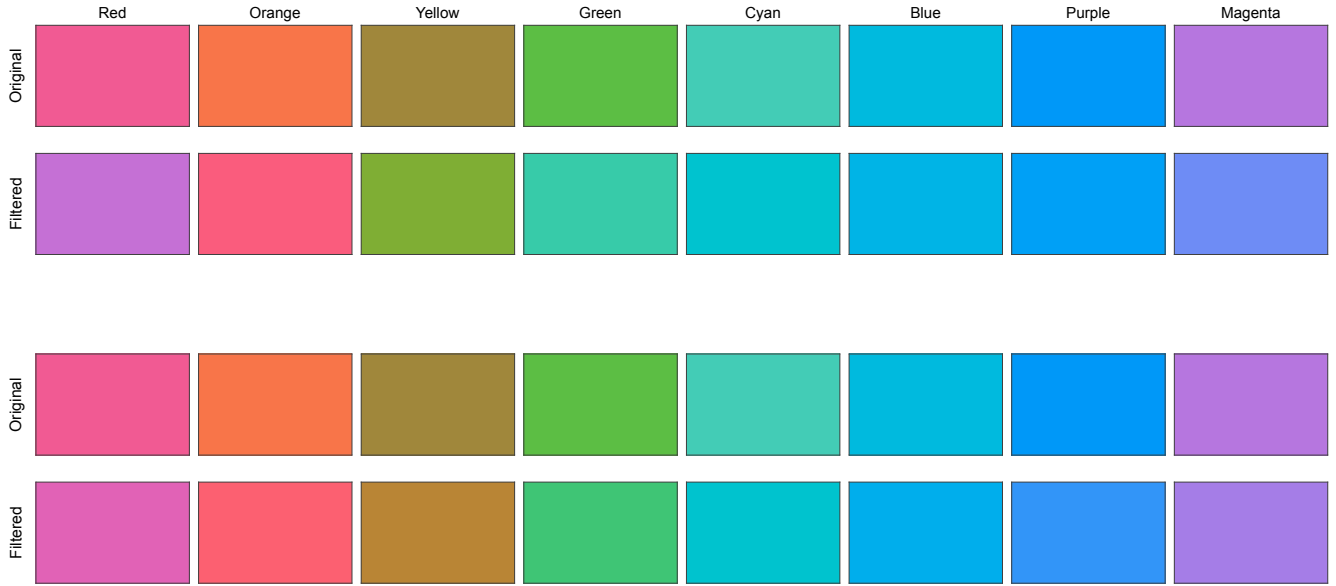


Figure 6: **Per-subject stimulus-space correction filter.** For each CVD participant, the displayed stimulus (*Original*) and the corrected stimulus (*Filtered*) are shown for the eight scanner hues, which run across the columns. Top block: the deutan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$. Bottom block: the protan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$. The filtered stimulus is the exact numerical pre-image of the fitted 2-component transform (8/8 pre-images exact, residual $< 0.001^\circ$ for both participants). The per-hue corrections $\delta\theta_{\text{filter}} = \theta_{\text{pre}} - \theta$ are reported in Section 3.6. Rendering coordinate space: STIM_LAB CIElab (project convention). Neural validation of both filters in a second session is shown in fig. 7. The psychophysical outcomes are reported in Section 3.7.

3.7 Psychophysical and neural filter evaluation

Psychophysics. The individualized filter lowered the mean threshold deviation from the controls in the deutan participant and left it unchanged in the protan participant, whereas the deployed filter lowered it in the deutan participant and raised it in the protan one. Session-1 baselines are reported in Section 3.3.

In the deutan participant both filters removed the baseline deficit. The three elevated pairs fell from +4.2, +4.3, and +6.7 control SDs to within ± 1.8 under both filters, and the mean $|z|$ fell from 2.24 to 0.85 under the deployed filter and 0.78 under the individualized one. The two filters were indistinguishable on thresholds (Wilcoxon $p = 0.84$).

In the protan participant the two filters diverged. The deployed filter raised the green–blue threshold from +2.4 to +5.1 control SDs ($p = 0.003$) and the cyan–magenta threshold from -0.3 to +6.2 ($p = 0.001$), lifting the mean $|z|$ from 0.90 to 1.78. The individualized filter lowered green–blue to +0.9 ($p = 0.44$), held every pair within ± 1.5 , and left the mean $|z|$ at 0.93. Only the individualized filter left both participants free of new deviant pairs.

Identification accuracy, held out from the fitting loss. In the protan participant, identification at ceiling without a filter (1.00, 95% CI [0.94, 1.00]) stayed preserved under the individualized filter (0.98, [0.92, 1.00]) and fell under the deployed comparator (0.86, [0.75, 0.92]). In the deutan participant, identification rose from 0.81 ([0.70, 0.89]) without a filter to 0.97 ([0.89, 0.99]) under each of the two filters. Eight-alternative color identification was held out of both fitting terms and therefore constitutes a prospective test independent of the loss from which the filters were derived.

Colors remained decodable. All eight hues remained decodable under both filters in both participants. LORO eight-way classification accuracy stayed above the 0.125 chance level at every ROI and in every condition, with the lowest cell at 0.50 against an HC range of 0.71 to 0.77 (Supplementary Table S18).

Interpolation. At hV4 the individualized filter raised interpolation in the deutan participant and lowered it in the protan participant. In the deutan participant, adjacent accuracy rose from 0.23 unfiltered to 0.31 under the individualized filter ($d_{cc} = -1.35$) and 0.25 under the deployed one ($d_{cc} = -1.94$), against an HC mean of 0.46 ± 0.11 , with the individualized value alone exceeding the 0.25 chance level. In the protan participant, the individualized filter gave the lowest of the three conditions (0.06, $d_{cc} = -3.70$, against 0.14 unfiltered and 0.19 deployed), and all three fell below chance. A secondary forward-tuning encoding index corroborated the deutan improvement at hV4 ($\rho = +0.18$ against HC $+0.21$, deployed $\rho = -0.39$) and did not separate the conditions in the protan participant ($\rho = -0.02$ in all three conditions; Fig. S3).

Geometry. Both filters left the representational geometry short of the HC level, and the direction of the filter effect reversed between the two participants (Fig. 7B,C,E,F; all values from the run-matched estimates of Supplementary §S15). In the deutan participant both filters moved the geometry further from HC than the unfiltered baseline: V2 SRM disparity rose from 0.68 to 0.87 deployed and 0.77 individualized (HC 0.44), and V2 RDM similarity fell from 0.42 to 0.16 and 0.05 (HC self-consistency 0.59). In the protan participant the two indices disagreed: V1 SRM disparity improved comparably under both filters (0.70 unfiltered to 0.66 deployed and 0.63 individualized; HC 0.43), whereas V1 RDM similarity rose only under the deployed filter (0.33 to 0.38) and fell under the individualized one (0.26; HC self-consistency 0.66). Neither index attributes the geometric recovery to individualization.

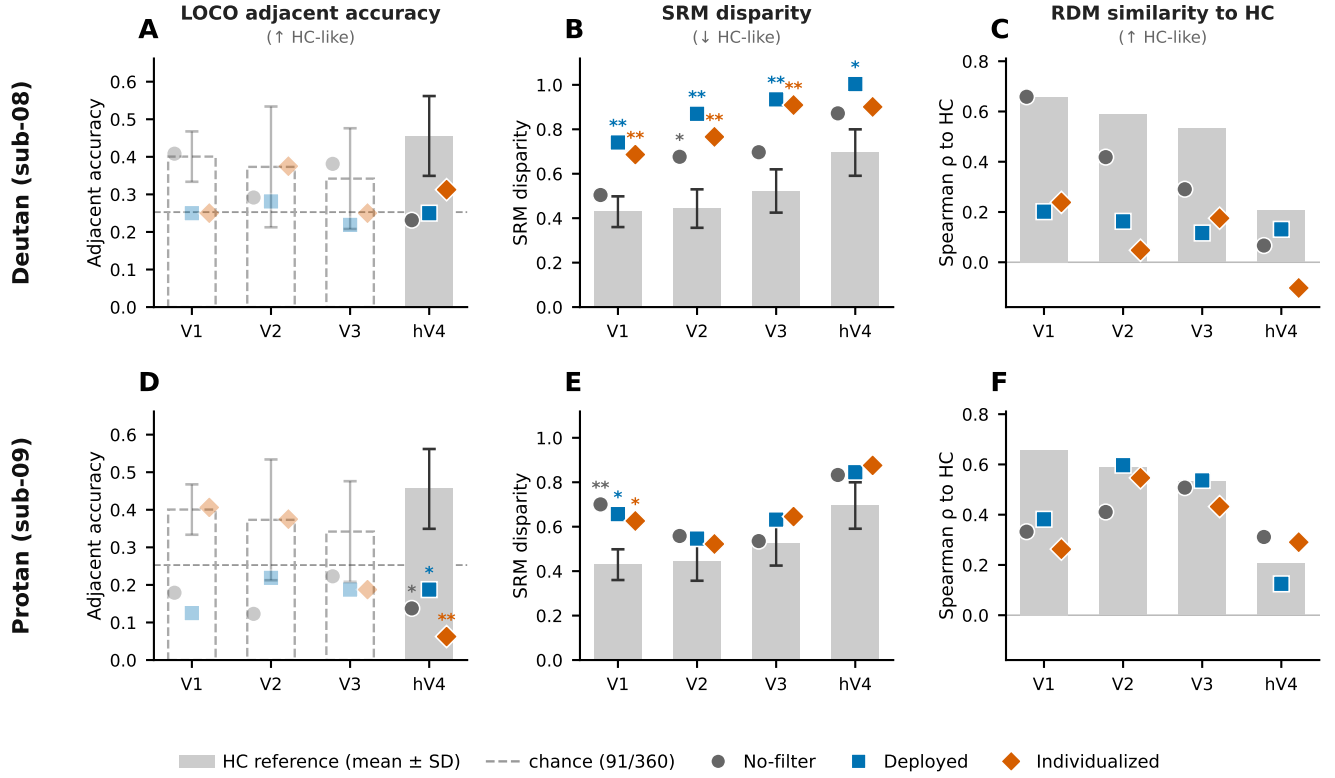


Figure 7: Hue interpolation and representational geometry under each filter in the second session. Rows: deutan (sub-08, A–C) and protan (sub-09, D–F). Columns: interpolation and representational geometry across V1–hV4. Markers: no-filter (gray circle, Session-1 baseline), deployed accessibility filter (blue square), individualized filter (orange diamond). Every panel is run-matched, with the HC reference and the unfiltered baseline built from four runs to match the filter conditions (Supplementary §S15). **(A, D)** LOCO adjacent accuracy, the proportion of predictions within one hue step of the target (↑ HC-like). Gray bars give the HC mean $\pm$ SD and the dashed line the analytic chance level of 0.25. hV4 is the region credited with interpolation (Section 3.1); V1–V3 are rendered muted with dashed outlines and are shown for comparison only, carrying no interpretation. **(B, E)** SRM disparity into the HC shared space (↓ HC-like). Gray bars give the HC mean $\pm$ SD. **(C, F)** Shared-space RDM similarity to HC (↑ HC-like). The gray bar is the HC leave-one-out self-consistency, a single value with no dispersion, so these panels carry no error bar and no significance marker. Stars in (A, B, D, E) give one-tailed Crawford–Howell single-case deviations from the HC distribution ($*p < .05$, $**p < .01$, $***p < .001$); in (A, D) they appear at hV4 only. The forward-tuning encoding index is in Fig. S3.

4 Discussion

In both CVD participants categorical color identification was preserved while continuous hue interpolation fell to chance at hV4. The hue geometry of both departed from the healthy-control (HC) reference by more than the controls departed from one another, and hue-discrimination thresholds were elevated on each participant's own confusion axis. We modeled that distortion as a hue rotation about two axes of color space, the subtype confusion axis and the S-cone axis, and fitted it to each participant's own neural and psychophysical measurements. We inverted each fitted model into a per-person stimulus-space filter and evaluated it in a second session. Under the individualized filter the deutan participant's baseline threshold deficit was removed and a trending confusion-axis pair in the protan participant was normalized, while every other pair stayed within the HC range. The deployed accessibility filter instead rendered two pairs deviant in the protan participant. On cortical hue interpolation the filter's effect varied by participant, with adjacent-hue accuracy increasing in the deutan participant and decreasing in the protan participant. Together these results locate the CVD deficit in continuous hue geometry, which categorical decoding does not detect. They also show that an individual's own cortical color representation can be inverted into a correction filter for that individual.

4.1 A geometric distortion of cortical color representation

Both CVD participants identified all eight hues well above chance at every region while their hue interpolation fell to chance at hV4, the only region where the controls interpolated above their color-label null. The preserved identification and the reduced interpolation each held under head-motion correction (Section 3.1; Supplementary §S1). Procrustes disparity from the control reference was elevated in both participants, while the region carrying the largest deviation was unstable across preprocessing in the deutan participant, leaving the cortical locus of the distortion undetermined (Section 3.2).

4.2 A neurally grounded, individualizable correction filter

The filter is the stimulus-space pre-image of the fitted distortion. For each intended color it substitutes the displayed hue that the participant's own cortical transformation maps onto that intended color (Methods §2.14). Each participant's loss combination was fixed beforehand by the pre-specified three-gate procedure over held-out control resamples (Methods §2.12). The two fitted corrections differed in magnitude and in direction, which parallels psychophysical evidence that hue perception varies markedly across anomalous trichromats (Emery et al., 2021). Their parameters come from each individual's own

measurements rather than from a subtype average, and the neural and psychophysical terms are not redundant. The neural term located a confusion-axis rotation that the psychophysical loss alone left near zero, and it narrowed the parameter uncertainty in both participants (Section 3.5). To our knowledge this is the first color-correction filter derived from an individual's cortical color representation rather than from a retinal model. Earlier inversions of cortical encoding models synthesized images that maximize the response of chosen units (Bashivan et al., 2019) or voxels (Shinkle & Lescroart, 2025) rather than the arrangement of colors within a region.

The fits recover the sign of the dominant confusion-axis term. No recovery check established the magnitude on either axis, and in the protan participant even that sign depends on the reduction basis and on the preprocessing pipeline (Supplementary §S14). With one participant per subtype, individual and subtype differences also remain confounded. Fixing those parameters therefore calls for a larger sample and a fitting procedure whose solution is stable across these analysis choices. This framework is built to support that analysis.

4.3 Filter evaluation

In the psychophysical evaluation both filters removed the deutan participant's baseline threshold deficit, whereas in the protan participant the individualized filter left every hue-discrimination pair within the HC range and preserved color identification accuracy while the deployed filter left two pairs deviant and lowered identification. The psychophysical advantage of the individualized filter therefore appeared in the protan participant only. The neural evaluation differed across measures. Hue interpolation at hV4 rose under the individualized filter in the deutan participant and fell in the protan participant, while the early-visual geometry moved in the opposite direction within each participant and did so under both filters alike, so neither shift was specific to the individualized correction. Representational similarity to the controls was higher under the deployed filter in both participants, and the geometry of both remained displaced from the HC reference under either filter (Section 3.7).

Early-visual geometry and hV4 interpolation lie at different levels of the visual hierarchy and can vary independently, so the disagreement alone does not show which readout tracks the correction. With one participant per subtype these neural endpoints remain inconclusive, and establishing whether either pattern generalizes requires replication in more individuals within each subtype on a refined neural endpoint.

4.4 Limitations

The CVD sample is $N = 2$, one participant per subtype. Subtype was assigned from Ishihara classification plates rather than anomaloscopy, so the confusion-axis angle assumed for each participant carries the uncertainty of that assignment. Single-case statistics (Crawford & Howell, 1998) permit individual-level inference, but population-level claims about deutan or protan subtypes require replication with more participants, and separating a per-person correction from a subtype-average one requires testing several individuals *within* a single subtype. The control cohort has to grow with it. At seven controls the leave-one-out distribution of HC $\|\hat{\beta}\|$ does not support a significance test, so we use it only to show where the CVD estimates fall relative to the control range.

Two of the reported estimates depend on analysis choices. The deutan V2 disparity elevation is significant in the common HC space and falls to a non-significant trend under the symmetric leave-one-subject-out control, whereas the protan V1 elevation is significant under both (Section 3.2). We also report $\hat{\beta}_s$ and $\hat{\beta}_c$ as single values without confidence intervals, because the six runs per participant are too few for stable bootstrap resampling.

The stimulus set was confined to a single locus of constant lightness and chroma ($L^* = 75$, chroma = 40). The correction is therefore untested for stimuli that vary in lightness or chroma, including natural scenes. Extending it to those stimuli requires refitting the model at additional levels of both dimensions. Lightness was equated in CIE $L^*a^*b^*$, which holds for the standard observer but not for an observer whose luminous efficiency departs from it, as it does in protanomaly. Measuring stimulus luminance and each participant's luminous efficiency would separate chromatic from luminance contributions to the cortical geometry.

The scanner task diverted attention from color. Participants monitored a letter stream at fixation while the colored disc was present, and no judgment about the disc was required. Attending to color raises response amplitude and tuning gain throughout visual cortex, and in ventral areas it additionally draws colors that share a name closer together (Brouwer & Heeger, 2013). The distortion reported here was measured with attention directed away from color, and a scanner task that requires a color judgment should resolve it more sharply.

The two neural terms in the fitting objective are not yet stable enough to combine. L_{LOCO} scores voxel-pattern prediction at hV4 and L_{RDM} scores pairwise distances at V1 or V2, so the two read different regions, and the region carrying the largest disparity itself moved with preprocessing in the deutan participant (Section 3.2). The interpolation term separated both CVD participants from the control distribution by a large margin, yet it entered neither selected combination under the three-gate

procedure. The same split reappears at evaluation, where adjacent accuracy, representational disparity, and RDM similarity to controls moved in different directions under the same filter and the individualized filter produced no consistent neural improvement (Section 3.7). Summing the two terms in a single objective therefore does not reconcile them. A neural endpoint that a larger study could act on would have to be stable across preprocessing, defined on a region fixed in advance, and shared between the fit and the evaluation.

4.5 Conclusion

All displayed hues remained decodable in both CVD participants, whereas their continuous hue geometry departed from the HC reference. In these two cases the distortion was a hue rotation about two axes of color space. We fitted that rotation to each participant's own cortical color representation and inverted it, which assigned every displayed hue a single replacement color and so yielded a stimulus-space correction for that participant. The fitted parameters differed between the two, so each correction is a per-person transform rather than a population-average retinal one.

Larger and systematic studies can now quantify the cortical color distortions of CVD and yield a class of individualized, cortically grounded filters. In the present two-person evaluation the correction's effect varied by participant and by measure. Whether it provides a generalizable improvement in color perception is what those studies would establish.

Declaration of the use of AI

With respect to the research described in this paper, AI tools played no part in the study design, in the choice of analyses, or in the interpretation of the results. AI-based coding assistants were used to review the analysis code and to check its correctness against the committed data. Every result reported here was produced by executing that code, and the authors verified the outputs.

With respect to the preparation of the manuscript, the authors used AI tools to check spelling and grammar and to reword draft text written by the authors. All statistical results and numerical values reported in the paper were taken from the analysis outputs rather than from an AI tool.

No figure contains AI-generated content. Every figure is produced by the analysis and figure scripts released with the paper.

The AI tools used were [FILL BEFORE SUBMISSION] exact tool names with version numbers, e.g. "Claude Code (Claude Opus 4.1)".

The authors take full responsibility for the content of this article, including any part prepared with the assistance of these tools.

Data and Code Availability

All analysis and preprocessing code needed to reproduce the reported results is openly available at https://github.com/Transconnectome/colorBlind_analysis [FILL AT SUBMISSION] and archived at Zenodo DOI xx.xxxx/zenodo.xxxxxxx. IN asks for a persistent identifier; make the repository public and mint a release DOI before submission so that reviewers can access it.

The neuroimaging and behavioural data cannot be shared openly. The protocol approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023), and the written consent obtained under it, place all study data in an encrypted store to which the principal investigator controls access, and neither document provides for secondary use by third parties or for deposit in a public repository. Under Article 16(3) of the Seoul National University Research Ethics Guidelines, the de-identified study data are retained for five years and, thereafter, for as long as the principal investigator can maintain custody of them, so that the integrity of the reported analyses can be verified. Requests for access should be directed to the principal investigator (Jiook Cha, connectome@snu.ac.kr) and are granted subject to (i) a formal data sharing agreement between the requesting institution and Seoul National

University, (ii) approval of the intended use by the requesting researcher's local ethics committee, and (iii) a short project outline describing that use; a request that falls outside the scope of the original consent additionally requires a new submission to the Seoul National University Institutional Review Board. Co-authorship is not a condition of access. Requests are answered within eight weeks.

Author Contributions

J.K.: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Project administration; Funding acquisition; Writing—original draft; and Writing—review & editing. A.M.C.: Conceptualization; Methodology; Formal analysis; Investigation; Data curation; Project administration; Funding acquisition; and Writing—review & editing. J.S.: Conceptualization (supporting); Investigation; Funding acquisition; and Writing—review & editing. J.C.: Conceptualization; Methodology; Validation; Resources; Supervision; Funding acquisition; and Writing—review & editing.

Ethics

The study was approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023) and was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants for being included in the study.

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Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. No patent has been filed on the correction filter described here, and no author has a financial relationship with a supplier of colour-correction eyewear or display software.

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Supplementary Methods

S1. Preprocessing pipelines and sensitivity analyses

The two pipelines. Every neural endpoint of the first session was computed under two pipelines. The primary pipeline resampled each functional volume once by a transform identical for every volume in a run. The second composed each volume's MCFLIRT rigid-body realignment matrix with that transform before the same single resampling. The two therefore differ in the alignment method and not in the number of resamplings. Head-motion correction lowered ROI temporal signal-to-noise ratio by 1.7 to 2.7%. In both, the per-run linear and constant drift terms of the general linear model were the whole of the nuisance model, and no temporal filter was applied. Slice timing and susceptibility distortion were left uncorrected in both. Within-run motion was estimated by registering every volume to its run's middle volume and summarized as framewise displacement (Power et al., 2012), with rotations converted to displacement on a 50 mm sphere. Across the nine analyzed participants mean framewise displacement was 0.318 ± 0.044 mm (HC 0.313 ± 0.042 , CVD 0.338 ± 0.046), and 16.2% of volumes exceeded 0.5 mm.

Disparity under the two pipelines. Procrustes disparity was recomputed under head-motion correction with every other element held fixed.

Table S1: Procrustes disparity by participant and region under the two pipelines. Entries give the Crawford–Howell one-tailed t with p in parentheses, against the healthy-control leave-one-out distribution ($n = 7$). The leave-one-subject-out estimator is the inferential one and is reported alongside it. The two primary-pipeline columns restate table S6, which adds the case-control effect sizes.

Participant	Region	Primary		Head-motion correction	
		Crawford–Howell	Leave-one-subject-out	Crawford–Howell	Leave-one-subject-out
Deutan	V1	1.1 (.157)	0.5 (.323)	2.4 (.027)	1.6 (.080)
Deutan	V2	2.1 (.040)	1.3 (.116)	−1.0 (.825)	−0.6 (.723)
Deutan	V3	1.9 (.052)	1.2 (.143)	0.6 (.293)	0.4 (.338)
Deutan	hV4	0.2 (.411)	0.1 (.474)	0.4 (.351)	0.4 (.356)
Protan	V1	3.5 (.007)	2.0 (.045)	1.6 (.077)	1.3 (.121)
Protan	V2	1.0 (.181)	0.8 (.234)	1.0 (.186)	0.1 (.480)
Protan	V3	0.1 (.466)	0.1 (.479)	1.1 (.151)	1.0 (.178)
Protan	hV4	1.1 (.150)	0.8 (.228)	1.4 (.101)	1.1 (.159)

The protan V1 elevation remained the largest deviation in that participant under both pipelines and weakened from $p = .007$ to $p = .077$. In the deutan participant the largest deviation moved from V2 to V1 and the V2 value reversed in sign. Under the leave-one-subject-out estimator the protan V1 cell of

the primary pipeline was the only one to reach significance. Individual cells of this grid are descriptive and support no claim in the main text.

Classification and interpolation under the two pipelines. The two readouts dissociate under both pipelines. The control-group interpolation gate, the two single-case comparisons at hV4 (§3.1), and the eight-way classification readout (§3.1) were recomputed the same way. hV4 alone passes the control gate in both pipelines, and both participants remain below the control mean in both. Classification is preserved in both. The lowest CVD cell across the four regions is 0.375 under the primary pipeline and 0.229 under head-motion correction, still 1.8 times the 0.125 chance level. No single-case classification contrast approaches significance in either pipeline ($p \geq .10$ in all sixteen cells).

Table S2: Hue-interpolation adjacent accuracy and eight-way classification accuracy under the two pipelines. Upper rows give the healthy-control interpolation mean with its p against 1,000 per-subject color-label permutations ($n = 7$; the permutation null mean is 0.35 in both pipelines). Middle rows give each CVD participant's interpolation accuracy at hV4 with the one-tailed Crawford–Howell p against the control distribution. Lower rows give classification accuracy at hV4, for which chance is 0.125 and the Crawford–Howell test is two-tailed because the hypothesis is preservation; the control mean is listed for reference and carries no test.

	Primary		Head-motion correction	
	Accuracy	p	Accuracy	p
<i>Interpolation (LOCO adjacent accuracy)</i>				
HC V1	0.393	.164	0.283	.922
HC V2	0.357	.424	0.381	.228
HC V3	0.339	.586	0.315	.810
HC hV4	0.456	.011	0.451	.023
Deutan hV4	0.250	.054	0.354	.242
Protan hV4	0.125	.012	0.271	.108
<i>Classification (LORO eight-way accuracy)</i>				
HC hV4	0.488	—	0.569	—
Deutan hV4	0.375	.351	0.583	.905
Protan hV4	0.375	.351	0.396	.197

Neither single-case interpolation contrast reached significance under head-motion correction. Individual cells of this grid are descriptive and support no claim in the main text.

Session-2 endpoints under the harmonized arm. We report both preprocessing arms for the second session and draw no directional conclusion from its cortical readouts under either. The session was reprocessed with the anatomical images harmonized to the first and with head-motion correction, and

the fourteen pre-specified endpoints were recomputed. The healthy-control reference and the Session-1 unfiltered anchor come from the same realigned reconstruction, so the two sessions are never mixed across arms. The control reference was essentially unchanged, with hV4 adjacent accuracy at 0.456 against 0.445. The single-participant, single-condition cells moved substantially. Eight of the ten directional contrasts defined on the native voxel mask reversed between arms, as did five of the ten defined on the run-matched mask. The two masks also disagree with each other. Each of those cells rests on four runs from one participant. The geometry indices moved less, and the four contrasts at the pre-specified deficit regions held in both arms. The control disparity reference itself shifts between arms, so only the ordering of conditions is comparable. The psychophysical endpoints do not depend on preprocessing and are unchanged.

Susceptibility distortion. Susceptibility distortion within the analyzed ROIs is predominantly subvoxel and spatially smooth, which makes it unlikely to account for the observed differences in representational geometry across color conditions. A gradient-echo field map was acquired in each session and linked to the functional runs through the BIDS IntendedFor field. Neither pipeline consumed it, so the data retain distortion along the right-left phase-encoding direction. Field-map-derived displacement was measured within each analyzed ROI in all nine participants, and mean displacement ranged from 0.01 to 0.76 voxels (0.02–1.52 mm). Within-ROI spatial variation is the component that distorts a pattern rather than translating it. It was 0.05–0.21 voxels (0.10–0.42 mm) in eight participants, with a 5th–95th percentile range of 0.15–0.72 voxels. In the ninth it reached 0.38 voxels (0.76 mm), with a range of 1.19 voxels. The remaining cost of the distortion is spatial, reducing the precision with which atlas-defined regions are assigned to functional voxels.

Registration. The functional series cover occipital cortex alone, so registration used mutual information initialized from the scanner-defined obliquity in the NIfTI header, which tolerates limited coverage and requires no white-matter boundary. The cost function was optimized to maximize the statistical dependence between the T1w and BOLD intensity distributions within the overlapping region. Two standard routes were evaluated first. A container-based route through fMRIPrep failed at the coregistration stage on every attempt with these partial-coverage series. Boundary-based registration within the custom pipeline was rejected on visual inspection. It snapped the functional slab onto an incorrect tissue boundary, an error on the order of 10 mm, where the mutual-information solution stayed within about 1 mm of the visually verified position. Whole-brain overlap indices favor boundary-based registration on these data. Dice and ROI-coverage measures respond to the extent of overlap rather than to the position of a partial slab within the brain, so we treat them as uninformative here and report the visual criterion

as the basis of the choice.

The mutual-information optimum is nonetheless shallow under this field of view. Across the runs of a single session, where the fitted transform should be identical, the mean pairwise displacement of the solution ranged from 0.9 to 4.2 mm. Removing facial voxels from the anatomical image, which leaves the brain unchanged, moved the solution by 1.9 mm in the deutan participant and 9.4 mm in the protan participant. The transform is fixed within a run, so it cannot produce differences among the eight colors that a run contributes. Registration variability instead enters as run-to-run noise, which averaging over the six runs attenuates. Rigid-body estimation and distortion correction are constrained just as weakly when little of the head is imaged, so acquisitions of this type would benefit from correction validated for partial coverage.

S2. Quality Control

ROI coverage averaged 83.5% (SD 22.7%) across the nine analyzed participants, and 99.5% of ROI voxels carried reliable stimulus-evoked responses. Coverage was computed as the intersection of the Wang atlas ROIs (V1, V2, V3, hV4) with each participant's BOLD brain mask in MNI space, taken per participant as the mean over the four regions and six runs. It reached 30.8% in sub-07, the lowest value in the sample. Every analyzed participant entered the downstream analyses.

S3. Mean Activation Analysis

Both CVD participants fell inside the control distribution on every activation metric at every ROI. Activation-level metrics were compared between the healthy controls and each CVD participant on the amplitudes before SRM alignment. The metrics were mean activation magnitude (mean $|\beta|$), signal-to-noise ratio, color modulation depth, run-to-run reliability, and spatial variance, each evaluated at V1, V2, V3 and hV4. Every comparison is a single-case test against the seven controls (Crawford and Howell (1998), two-tailed), the procedure used for the geometric and interpolation endpoints.

For mean $|\beta|$ every value satisfied $p \geq 0.182$, the largest deviation being $d_{cc} = +1.61$ in the deutan participant at V3. For color modulation depth the largest deviation was $d_{cc} = +2.16$ in the deutan participant at V2 ($p = 0.090$). The largest deviations on both metrics were positive rather than negative, which is the opposite of the direction a uniform attenuation predicts. Voxel-level color selectivity was substantial in both groups (one-way ANOVA across voxels, $F > 4$, $p < 0.001$ in V1–V3). With seven controls a two-tailed test at $\alpha = 0.05$ requires a deviation of 2.62 control standard deviations, and its

power against a true deviation of 2 SD is approximately 0.35. These tests therefore rule out a substantial uniform attenuation rather than establishing equality.

The HC–CVD difference resides in the representational pattern rather than in response amplitude. Activation metrics and SRM disparity varied independently across participants, with Pearson r from -0.29 to $+0.51$ over 20 tests and all $p \geq 0.130$.

S4. Dimensionality Selection (K-Selection)

SRM used $k = 4$ at V1 and V2 and $k = 3$ at V3 and hV4, with cross-subject RDM Spearman correlations of 0.60, 0.57, 0.55 and 0.32. The reduced dimension was determined by leave-one-subject-out cross-validation over the candidate values 2 to 6 (Cunningham & Yu, 2014). On each of the seven folds, one HC participant was withheld from SRM training and three quality metrics were computed for every candidate. RDM split-half reliability was the Spearman correlation between representational dissimilarity matrices built from the odd runs 1, 3, and 5 and the even runs 2, 4, and 6 of the withheld participant in SRM space. The remaining two metrics were the cross-subject RDM correlation and the reconstruction error. Candidates were ranked within each fold and metric, and the candidate with the best mean rank was selected.

S5. Generalized Cross-Validation for Ridge Regression

The ridge penalty applies to the encoding direction alone. Hue decoding uses the unregularized pseudoinverse, because a ridge penalty on the channel-to-voxel weights degrades the decoding readout (§S7). The regularization parameter α was selected by generalized cross-validation (Golub et al., 1979), which approximates leave-one-out cross-validation at a fraction of its cost. We implemented it directly from the singular value decomposition of the channel design matrix, searching the grid $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$. A single α was selected per fit by minimizing the GCV score,

$$\text{GCV}(\alpha) = \frac{\overline{\|Y - \hat{Y}\|_2^2}}{(1 - \text{tr}(H)/n)^2} \quad (5)$$

where C denotes the channel design matrix, Y the observed responses of all voxels in the region, $\hat{Y}$ the ridge prediction, $H = C(C^T C + \alpha I)^{-1} C^T$ the hat matrix, and n the number of training samples. The squared residuals are averaged over both samples and voxels, so the selected penalty is shared across the voxels of a region.

S6. Cross-validation procedures and evaluation metrics

The two within-subject schemes are defined in Methods §2.8. Leave-one-run-out (LORO) holds out one run at a time and keeps every hue in training, and leave-one-color-out (LOCO) holds out all six runs of one hue and keeps that hue out of training. Both decode the held-out patterns with the OLS pseudoinverse weights of Methods §2.7. The complementary voxel-response prediction by ridge-GCV encoding enters the fitting loss L_{LOCO} alone and never the decoding metric. This section reports the fold-level checks and the evaluation metrics that the main text does not carry.

Cross-run alignment and leakage control. The run-level Procrustes alignment (Methods, ROI definition and response estimation) maps runs 2–6 onto a fixed run-1 reference by an orthogonal rotation that uses no stimulus labels. Because the alignment target is a single fixed run rather than a jointly fitted shared space, the held-out run or color does not define the alignment frame. Re-estimating the alignment inside each fold raised forward-encoding LORO accuracy from 0.545 to 0.578. Leakage in the fixed-reference pipeline would predict the opposite direction, so the reported estimate is not inflated by the alignment step. That variant also replaces the run-1 reference with the training-run mean, so it bounds the leakage question rather than isolating the nesting itself.

Leave-one-subject-out (LOSO). LOSO evaluated zero-shot generalization to held-out participants. Cross-subject prediction needs a common voxel space, so LOSO ran on the SRM-transformed data. For each HC participant, W was trained on the remaining HC participants' shared-space data and tested on the held-out participant.

Color decoding metrics. Color decoding was evaluated on two metrics. The first is mean absolute error between the predicted and the true hue angle, for which a uniform random predictor on a circle gives a chance level of 90°. The second is eight-way classification accuracy, obtained by assigning the decoded hue to the nearest stimulus hue, which is a $\pm 22.5^\circ$ bin around each of the eight colors and gives a chance level of 12.5%.

Voxel prediction metrics. Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

1161 S7. Comparison with alternative decoders

1162 The forward encoding model was adopted for its interpolation behavior rather than for its classification
 1163 accuracy. We compared six decoders on the same SRM-aligned amplitudes under both cross-validation
 1164 schemes, which separate them in opposite directions. The interpolation result is the one the present
 1165 analyses depend on.

1166 Linear discriminant analysis classified held-out runs most accurately at every ROI (table S3). Its healthy-
 1167 control mean reached 0.878 at V1 against a chance level of 0.125, with the support vector machine next
 1168 at 0.777 and the forward encoding model at 0.542. The ordering held at V2, V3 and hV4. Four of the
 1169 six decoders exceeded chance at every ROI, so eight-way classification is recoverable by several readouts
 1170 and does not by itself favor any one of them.

1171 Interpolation to a held-out color reverses the ordering at hV4, the ROI on which the main analyses rest.
 1172 The forward encoding model reached an adjacent accuracy of 0.470 there, against a chance level of 0.25
 1173 for its continuous-hue output (table S4). It is the only decoder whose healthy-control mean exceeds its
 1174 own chance level, doing so at V1 (0.360), V2 (0.283) and hV4, with the widest margin at hV4. The
 1175 classifiers stay below their 0.375 chance level at every ROI, and ridge and kernel ridge stay below 0.25
 1176 everywhere. The analytic chance level compares a decoder against a uniform prediction. The stricter
 1177 main-text criterion uses a color-label permutation null that preserves the covariance among voxels, and
 1178 hV4 is the region that meets it (§3.1). The pattern follows from the model form. The forward encoding
 1179 model represents a hue as activation over six tuned channels and can therefore evaluate a hue it never
 1180 saw during training, whereas a classifier trained on seven discrete labels has no output unit for the eighth.

1181 The correlation readout entered this comparison as the incumbent rather than as its outcome. It was
 1182 implemented for the Phase-1 forward model in January 2026 and specified in the analysis plan in February
 1183 2026, both before the runs reported here. The comparison therefore verifies that the readout was not
 1184 the limiting factor, and the wording in the main text reflects that scope.

1185 Two rows require care. Kernel ridge regression returned an adjacent accuracy of exactly 0.000 in every
 1186 participant, ROI and alignment, and the multilayer perceptron returned exactly the chance value with
 1187 zero variance across participants, which is the signature of a constant prediction. Neither is treated as
 1188 a competitive score. Three further variants sit outside this table. The ensemble decoders were retired
 1189 from the pipeline, the two hybrid decoders were run for interpolation only, and four alternative readouts
 1190 of the forward encoding model vary the readout rather than the decoder.

Table S3: Eight-way classification accuracy under leave-one-run-out cross-validation, SRM-aligned amplitudes, chance = 0.125. Healthy-control mean $\pm$ SD ($n = 7$), with the two CVD participants listed separately. Bold: highest control mean in the ROI.

ROI	Decoder	HC (mean $\pm$ SD)	Deutan	Protan
V1	LDA	0.878 $\pm$ 0.050	1.000	0.854
	SVM	0.777 $\pm$ 0.062	0.979	0.771
	Forward encoding	0.542 $\pm$ 0.106	0.604	0.625
	Ridge	0.399 $\pm$ 0.054	0.354	0.250
	Kernel ridge	0.393 $\pm$ 0.064	0.271	0.292
	MLP	0.128 $\pm$ 0.008	0.125	0.125
V2	LDA	0.830 $\pm$ 0.047	0.917	0.854
	SVM	0.759 $\pm$ 0.136	0.875	0.833
	Forward encoding	0.545 $\pm$ 0.077	0.458	0.438
	Ridge	0.372 $\pm$ 0.097	0.458	0.250
	Kernel ridge	0.330 $\pm$ 0.066	0.313	0.250
	MLP	0.149 $\pm$ 0.031	0.125	0.125
V3	LDA	0.726 $\pm$ 0.064	0.750	0.771
	SVM	0.631 $\pm$ 0.144	0.813	0.667
	Forward encoding	0.449 $\pm$ 0.065	0.375	0.313
	Ridge	0.173 $\pm$ 0.068	0.292	0.229
	Kernel ridge	0.158 $\pm$ 0.052	0.208	0.333
	MLP	0.122 $\pm$ 0.008	0.125	0.125
hV4	LDA	0.658 $\pm$ 0.092	0.813	0.729
	SVM	0.598 $\pm$ 0.085	0.813	0.667
	Forward encoding	0.423 $\pm$ 0.072	0.354	0.354
	Ridge	0.319 $\pm$ 0.114	0.146	0.250
	Kernel ridge	0.277 $\pm$ 0.093	0.042	0.188
	MLP	0.125 $\pm$ 0.000	0.125	0.125

S8. Alignment Robustness for the Within-Subject Readouts

The two alignment spaces agree on the pattern that carries the Results. LORO and LOCO are trained and tested within a single participant, so the main text computes them on the Procrustes-aligned amplitudes, which retain that participant's voxels at full resolution. SRM is applied only where a comparison spans participants. table S5 repeats both readouts in the SRM-aligned space as a robustness check.

Classification stays far above the 0.125 chance level for both CVD participants at every region in both spaces, with the lowest cell at 0.312. Interpolation at hV4 remains reduced in both participants in both spaces, and the ordering of the two participants is preserved. Procrustes alignment yields the higher control-mean accuracy at all four regions for classification and at three of the four for interpolation. It

Table S4: Adjacent accuracy under leave-one-color-out cross-validation, SRM-aligned amplitudes. Healthy-control mean $\pm$ SD ($n = 7$), with the two CVD participants listed separately. Chance depends on the output space of the decoder. Forward encoding, ridge and kernel ridge return a continuous hue, so a uniform prediction lands within the 45° tolerance on 91 of 360 draws and chance is 0.25. The LDA, SVM and MLP classifiers return one of the eight stimulus hues, for which the tolerance admits 3 of 8 and chance is 0.375. The constant-output MLP sits at exactly 0.375 at V1 and V2, which confirms the classifier value. Bold: the largest margin above the applicable chance level. Kernel ridge and the MLP are constant-output rows, noted in the text.

ROI	Decoder	HC (mean $\pm$ SD)	Deutan	Protan
V1	Forward encoding	0.360 ± 0.098	0.313	0.250
	LDA	0.268 ± 0.074	0.375	0.167
	SVM	0.226 ± 0.078	0.375	0.229
	Ridge	0.039 ± 0.053	0.146	0.000
	MLP	0.375 ± 0.012	0.375	0.375
	Kernel ridge	0.000 ± 0.000	0.000	0.000
V2	Forward encoding	0.283 ± 0.101	0.333	0.063
	LDA	0.327 ± 0.088	0.313	0.417
	SVM	0.318 ± 0.108	0.250	0.396
	Ridge	0.054 ± 0.083	0.021	0.000
	MLP	0.375 ± 0.000	0.396	0.375
	Kernel ridge	0.000 ± 0.000	0.000	0.000
V3	Forward encoding	0.220 ± 0.100	0.354	0.125
	LDA	0.220 ± 0.081	0.271	0.333
	SVM	0.208 ± 0.103	0.250	0.229
	Ridge	0.000 ± 0.000	0.000	0.000
	MLP	0.250 ± 0.000	0.250	0.250
	Kernel ridge	0.000 ± 0.000	0.000	0.000
hV4	Forward encoding	0.470 ± 0.130	0.271	0.104
	LDA	0.280 ± 0.144	0.146	0.229
	SVM	0.253 ± 0.126	0.167	0.208
	Ridge	0.000 ± 0.000	0.000	0.000
	MLP	0.250 ± 0.000	0.250	0.250
	Kernel ridge	0.000 ± 0.000	0.000	0.000

is also the higher of the two spaces in 26 of the 36 participant-by-region classification cells. This is consistent with the dimensionality reduction in SRM discarding response variance that a within-subject readout can otherwise use.

The spaces differ at V1 classification, where SRM places both CVD participants above the control mean ($d_{cc} = +0.59$ and $+0.79$) while Procrustes places them slightly below ($d_{cc} = -0.25$ for both). No single-case classification comparison reaches significance in either space.

Table S5: Forward-encoding readouts in the two alignment spaces. LORO is eight-way classification accuracy (chance 0.125); LOCO is adjacent accuracy (analytic chance 0.25). HC is the mean over $n = 7$. The main text reports the Procrustes columns.

Region	LORO classification			LOCO interpolation		
	HC	deutan	protan	HC	deutan	protan
<i>Procrustes-aligned (main text)</i>						
V1	0.580	0.562	0.562	0.393	0.437	0.188
V2	0.607	0.521	0.562	0.357	0.271	0.104
V3	0.574	0.396	0.458	0.339	0.375	0.208
hV4	0.488	0.375	0.375	0.456	0.250	0.125
<i>SRM-aligned (robustness)</i>						
V1	0.542	0.604	0.625	0.360	0.312	0.250
V2	0.545	0.458	0.438	0.283	0.333	0.062
V3	0.449	0.375	0.312	0.220	0.354	0.125
hV4	0.423	0.354	0.354	0.470	0.271	0.104

S9. Leave-One-Out Consistent Disparity Estimation

Disparity was estimated against leave-one-HC-out references. For each fold i the reference R_{-i} is the mean of the remaining six HC participants' aligned data. The held-out HC_i and both CVD participants are compared against that same reference, so HC and CVD disparities are measured against identical references within each fold. Individual CVD significance used the Crawford and Howell (1998) modified t , comparing each CVD participant's mean disparity over the seven folds to the HC distribution ($n = 7$, $df = 6$).

Common-space vs. symmetric-LOSO estimates. The geometric-disparity tests (§3.2) are reported primarily in the common HC-trained SRM space, in which the seven HC train the shared space and each subject of interest is projected in. In that space the held-out HC reference is in-sample whereas the projected CVD participant is out-of-sample. We therefore computed a symmetric leave-one-subject-out (LOSO) estimate as well, in which every HC and CVD subject alike is projected by SVD into a space trained on the remaining six HC (Methods, §2.9). Both use the Crawford and Howell, 1998 one-tailed single-case t ($df = 6$), and Table S6 reports both for all ROIs. The protan participant's V1 elevation is significant under both estimators. The deutran participant's V2 elevation is significant in the common space and does not survive under LOSO ($p = 0.116$, $d_{cc} = 1.42$). V3 in that participant behaves the same way under both estimators ($p = 0.052$ and $p = 0.143$), so the two regions are not separable in the deutran participant.

Table S6: Procrustes disparity, Crawford–Howell one-tailed single-case t ($df = 6$) against the HC reference, in the primary common HC-trained SRM space and under the symmetric LOSO control, with the case-control effect size $d_{cc} = t\sqrt{8/7}$ ($n = 7$) given for each estimator. Bold: $p < 0.05$.

Subject	ROI	Common space			Symmetric LOSO		
		t	p	d_{cc}	t	p	d_{cc}
Deutan	V1	1.10	0.157	1.18	0.48	0.323	0.51
	V2	2.11	0.040	2.26	1.33	0.116	1.42
	V3	1.92	0.052	2.05	1.17	0.143	1.25
	hV4	0.23	0.411	0.25	0.07	0.474	0.07
Protan	V1	3.48	0.007	3.72	2.02	0.045	2.16
	V2	0.99	0.181	1.06	0.77	0.234	0.82
	V3	0.09	0.466	0.10	0.06	0.479	0.06
	hV4	1.13	0.150	1.21	0.80	0.228	0.86

S10. Validity of the geometric comparison

The color-correspondence permutation carries power only when the shared projection is held fixed, since re-estimating it at each permutation lets the re-fit absorb the shuffle. The permutation asks whether a participant's colors map onto the healthy-control geometry in the identity order. It differs from the color-label permutation null of the main text, which asks whether the control group interpolates above chance in a region. We tested both variants on the healthy controls, in whom color structure is established. Re-estimation detected 0 of 7 at V1, V2 and V3 and 0 of 6 at hV4, with mean permutation z near zero in every ROI. Freezing the projection recovered detection, reaching 5 of 7 at V3 and 2 to 3 of 7 elsewhere (table S7). Every color-specificity result below uses the frozen projection.

Table S7: Color-correspondence permutation on the healthy controls, used as a positive control. Detection counts and mean permutation z under a projection re-estimated at each permutation and under a frozen projection. Lower z indicates a better identity correspondence than the shuffled null.

ROI	n	Re-estimated		Frozen	
		detected	mean z	detected	mean z
V1	7	0	−0.04	2	−1.58
V2	7	0	−0.04	3	−1.00
V3	7	0	−0.78	5	−2.39
hV4	6	0	+0.06	2	−1.07

Both CVD participants carried color-specific geometry. The deutan participant reached $p = .002$ at V2, the lowest V2 value among the nine participants, and $p = .024$ at V3. The protan participant reached $p = .001$ at V3 and $p = .013$ at V2, while V1 returned $p = .758$. Across the 35 participant-by-ROI cells,

16 fell below .05 against a chance expectation of 1.8, and 7 survived Benjamini–Hochberg correction over all 35 (table S8). Two of the survivors belong to the CVD participants, the deutan V2 cell ($q = .018$) and the protan V3 cell ($q = .012$). V3 carried color specificity most broadly, with 5 of its 9 cells surviving.

Table S8: Color-correspondence permutation under the frozen projection, one cell per participant and region, in the two preprocessing pipelines. Entries give p_{perm} against $N = 1,000$ permutations. Seven cells survive Benjamini–Hochberg correction across the 35 cells of the primary pipeline and none survive it in the head-motion-corrected pipeline.

Participant	Primary				Head-motion correction			
	V1	V2	V3	hV4	V1	V2	V3	hV4
HC 1	0.088	0.020	0.062	0.393	0.012	0.072	0.018	0.290
HC 2	0.001	0.016	0.004	0.136	0.437	0.420	0.191	0.019
HC 3	0.109	0.520	0.255	0.272	0.325	0.113	0.176	0.101
HC 4	0.152	0.014	0.001	0.041	0.420	0.006	0.015	0.173
HC 5	0.057	0.773	0.034	0.031	0.079	0.194	0.032	0.038
HC 6	0.407	0.290	0.009	0.220	0.373	0.117	0.010	0.012
HC 7	0.034	0.159	0.004	—	0.008	0.005	0.391	—
Deutan	0.105	0.002	0.024	0.273	0.022	0.003	0.047	0.257
Protan	0.758	0.013	0.001	0.129	0.010	0.148	0.053	0.815

Under head-motion correction 15 cells fell below .05 and none survived correction. The deutan V2 cell held its nominal value across the two pipelines ($p = .002$ and $p = .003$) while its corrected value moved from $q = .018$ to $q = .053$. The protan V1 cell moved from $p = .758$ to $p = .010$. Individual cells of this grid are descriptive and support no claim in the main text.

Disparity and color specificity address different questions. Disparity measures the distance between two geometries after optimal rotation and stays invariant to how the colors map onto one another. The permutation asks whether the identity mapping between a participant’s colors and the reference colors outperforms a shuffled mapping. A participant can therefore sit far from the reference in disparity while the identity mapping remains the best correspondence available, or sit close to it while a shifted correspondence fits better.

The protan participant’s V1 correspondence is displaced by one hue step, and the failure is not one of signal. That participant’s V1 split-half pattern reliability reaches .847, above the highest control value, and eight-way classification there reaches 0.79 against a chance level of 0.125. Rotating the color labels by 45° lowered Procrustes disparity from 1.037 to 0.788, a gain of 24.0%. We referred that gain to the distribution of shift gains across the seven healthy controls ($3.5 \pm 5.9\%$), which is computed under the same minimum-over-eight rule and therefore carries the same selection bias. The protan gain exceeded that distribution (Crawford–Howell $t = 3.22$, $p = .009$, $d_{cc} = 3.44$), and the rotated geometry sat below

the control mean disparity of 0.839 ± 0.087 . The same rotation raised the second-order RSA similarity from $\rho \approx 0.00$ to $\rho = +0.52$, against a control mean of 0.45, exceeding its own selection-bias null ($z = +5.02$, $p = .002$, $p_{\text{adj}} = .032$). Disparity at this region is nonetheless elevated in both pipelines, so the departure is not accounted for by a rigid relabeling alone. The account applies to the primary pipeline, since that cell is no longer null under head-motion correction. The deutan participant showed the identity mapping as optimal at V1, V2 and V3.

Which rearrangement restores control-level similarity remains open. Under the SRM basis a 225° shift ($+0.50$) sits close to the 315° optimum ($+0.52$). Under the PCA basis the optimum falls at 135° ($p = .048$, which does not survive correction).

S11. Alignment-independent checks on the disparity measure

Disparity tracks the same subject-level quantity when the alignment method changes. We recomputed each participant's distance from the healthy-control mean with two procedures that use no shared response model, then correlated the result with the SRM disparity used in the main analyses. Crossnobis distance in native voxel space agreed at $r = 0.632$ pooled across ROIs ($p < .001$) and PCA distance at $r = 0.780$ ($p < .001$). Agreement was strongest at V1 and V2, the two ROIs carrying the single-case elevations (table S9).

Cross-validated Mahalanobis distance reproduces the ordering without any alignment step. Following Walther et al. (2016), we computed an 8×8 crossnobis distance matrix per participant in native voxel space, cross-validated over the 15 run pairs, with Ledoit–Wolf shrinkage of the noise covariance. Control-to-control RDM similarity exceeded control-to-CVD similarity at V1 by 0.104 (permutation $p = .120$), and the three remaining ROIs gave differences below 0.05. The convergent correlation with SRM disparity reached $r = 0.833$ at V1 ($p = .005$) and $r = 0.733$ at V2 ($p = .025$).

A second alignment algorithm gives the same subject ordering. We replaced SRM with PCA at matched dimensionality, with and without a subsequent canonical-correlation alignment, and computed Procrustes disparity over all participant pairs. Group-level effects were small under both variants, with Hedges' g from -0.13 to $+0.40$ under PCA and from -0.13 to $+0.16$ under PCA-CCA. Pairwise alignment fits each pair independently and therefore carries more noise than a jointly fitted shared space, which accounts for their size. Subject-level agreement with SRM disparity was substantial under both (PCA pooled $r = 0.780$, $p < .001$; PCA-CCA pooled $r = 0.544$, $p < .001$).

The shared space reconstructs the CVD data at least as well as the control data, which places re-

construction quality outside the set of explanations for the elevated disparity. Under the symmetric leave-one-subject-out framework, variance explained was higher in CVD than in controls at all four ROIs, and the V2 difference was the largest (controls 0.331, CVD 0.416, $g = -1.26$). Variance explained and disparity varied independently across participants (pooled $r = -0.214$, $p = .211$), so the two indices track separate properties. High reconstruction quality together with high disparity describes a strong signal held in a different arrangement.

These checks concern the measurement alone. Whether the deviation is specific to color is treated in §S10. Group-level contrasts at this sample size carry wide intervals, so the argument rests on the subject-level agreement between three distances computed under three different alignment procedures.

Table S9: Convergent validity of SRM disparity with two alignment-independent distances, Spearman r over the nine analyzed participants. Crossnobis distance uses native voxel space with no alignment step. PCA and PCA-CCA replace the shared response model with a pairwise alignment at matched dimensionality. Bold: $p < 0.05$.

Distance	V1	V2	V3	hV4	Pooled
Crossnobis (voxel space)	0.833	0.733	0.550	0.300	0.632
PCA	0.633	0.850	0.333	0.533	0.780
PCA-CCA	0.483	0.433	0.100	-0.067	0.544

Table S10: Variance explained by the shared response model under the symmetric leave-one-subject-out projection, in which controls and CVD participants are projected identically. Hedges' g is signed control minus CVD, so negative values indicate better reconstruction of the CVD data.

ROI	k	Controls	CVD	g
V1	4	0.352	0.407	-0.40
V2	4	0.331	0.416	-1.26
V3	3	0.250	0.314	-0.66
hV4	3	0.225	0.253	-0.39

S12. Session-1 Hue-Discrimination Thresholds

table S11 lists the unfiltered Session-1 threshold for every tested hue pair. The threshold is the staircase level t , the fraction of the pair's full CIE LCh separation at which the two discs were discriminated (§2.3), so smaller values indicate finer discrimination. The ratio γ divides the participant's threshold by the healthy-control mean, and z expresses the same difference in control standard deviations over the seven controls. Pairs are grouped by the axis each was chosen to probe, and that assignment was fixed before any threshold was collected.

The elevations align with subtype. The deutan participant exceeded $z = +4$ on both deutan-axis pairs and on the S-cone pair, whereas the protan participant exceeded the control range only on the protan-axis pair green–blue. Neither participant was elevated on the 180° control pair, which argues against a general loss of sensitivity. These are the values entering the psychophysical loss atoms of §2.10. Single-case tests of the same pairs are reported with the filter evaluation, where they serve as the pre-filter baseline (§3.7).

Table S11: Session-1 hue-discrimination thresholds without a filter. HC is the mean $\pm$ SD over the seven healthy controls. γ is the ratio to the HC mean and z the deviation in HC standard deviations. Bold: $|z| \geq 2$. Two staircases returned an incorrect response at the largest presentable separation, both of them the deutan participant’s orange–yellow pair, so that threshold is a lower bound rather than an estimate. It is the only such pair among the 208 staircases collected in this study.

Axis probed	Pair	HC ($n = 7$)		Deutan			Protan		
		mean	SD	t	γ	z	t	γ	z
Deutan confusion	orange–yellow	0.278	0.135	0.840	3.02	+4.15	0.193	0.69	−0.63
	yellow–green	0.090	0.044	0.278	3.10	+4.32	0.128	1.42	+0.87
Protan confusion	green–blue	0.079	0.032	0.078	0.98	−0.06	0.155	1.95	+2.36
	cyan–magenta	0.042	0.017	0.040	0.95	−0.13	0.038	0.89	−0.28
S-cone	yellow–purple	0.022	0.006	0.063	2.87	+6.70	0.028	1.26	+0.94
	blue–purple	0.164	0.093	0.120	0.73	−0.48	0.148	0.90	−0.18
	red–orange	0.124	0.072	0.063	0.50	−0.85	0.075	0.61	−0.68
Control	red–cyan	0.034	0.015	0.015	0.45	−1.23	0.015	0.45	−1.23
Mean $ z $ over the eight pairs		Deutan 2.24		Protan 0.90					

The deutan orange–yellow threshold is censored rather than estimated, and the true value lies at or above the tabulated one. Both staircases of that pair spent their whole course at the top of the presented range and returned incorrect responses at the largest separation the task can present. The censoring understates the baseline deficit and, with it, the improvement recorded under either filter, so the values are reported unadjusted.

table S12 gives both staircase estimates for every pair and condition. Across all participants and conditions the two staircases of a pair differed by a median of 0.015 in 104 pairs. Three cells exceeded 0.10, two of them controls, a control’s red–orange pair at 0.190 and another control’s orange–yellow pair at 0.130. The largest was the protan participant’s orange–yellow pair under the individualized filter, at 0.515. One staircase there settled at 0.715 after having answered correctly at a separation four times smaller earlier in the same block. Its partner converged normally at 0.200 on the same rendered stimuli. A rendering failure would have disturbed both members of the pair, and the value lies well below the largest presentable separation of 0.95, so neither rendering nor censoring accounts for it. A single lapse on one track does.

Table S12: Both staircase estimates for every color pair, given as high-start/low-start in units of the rendered separation. Bold marks the cells whose two estimates differ by more than 0.10. Thresholds reported elsewhere are the mean of the two.

Pair	Deutan			Protan		
	Session 1	Deployed	Individualized	Session 1	Deployed	Individualized
red–orange	0.060/0.065	0.160/0.210	0.140/0.130	0.075/0.075	0.110/0.145	0.070/0.055
orange–yellow	0.870/0.810	0.080/0.110	0.060/0.032	0.175/0.210	0.170/0.185	0.715/0.200
yellow–green	0.280/0.275	0.090/0.075	0.145/0.165	0.125/0.130	0.057/0.050	0.055/0.065
green–blue	0.065/0.090	0.150/0.110	0.045/0.060	0.145/0.165	0.260/0.225	0.135/0.080
blue–purple	0.115/0.125	0.050/0.095	0.110/0.200	0.125/0.170	0.135/0.110	0.150/0.105
yellow–purple	0.060/0.065	0.015/0.020	0.035/0.025	0.020/0.035	0.020/0.015	0.015/0.015
red–cyan	0.015/0.015	0.040/0.018	0.015/0.045	0.015/0.015	0.030/0.040	0.020/0.025
cyan–magenta	0.035/0.045	0.030/0.025	0.040/0.033	0.040/0.035	0.150/0.140	0.015/0.020

Orange–yellow carries the largest threshold of the eight pairs in nine of the thirteen participant-by-condition blocks, which is where a staircase has the most room to be destabilized by one response. A separate cell of the same pair, the deutan participant’s session-1 measurement at 0.870/0.810, is instead a censoring case and is treated in the preceding paragraph. All thresholds are reported as the mean of the two staircases without adjustment. For the protan orange–yellow cell, using the converged staircase alone would move it from $z = +1.33$ to $z = -0.58$ and the participant’s mean $|z|$ from 0.93 to 0.84.

S13. Comparison with Retinal-Family Distortion Models

The retinal-family (R+C) model does not represent the fitted distortion and saturated its gain boundary in both participants. It was fit by the same grid search and held-out composite criterion as the 2-component model (§2.12), with g searched over $[0, 3]$ in steps of 0.05, so the two classes are directly comparable on $\bar{L}_{\text{test}}$.

Model. The prevailing account attributes CVD color distortion to a cone-spectral shift at the retinal level (Machado et al., 2009), optionally augmented by a cortical compensation gain g (Boehm et al., 2014; Tregillus et al., 2021):

$$\delta\theta_{\text{RC}}(\theta; g) = (2 - g) \delta\theta_{\text{Mach}}(\theta) \quad (6)$$

where $\delta\theta_{\text{Mach}}$ is the per-hue angular shift predicted by the Machado model. That model is defined on the Smith and Pokorny cone fundamentals, and we evaluated it on the Stockman and Sharpe fundamentals instead (Stockman & Sharpe, 2000). The gain g parameterizes cortical compensation. At $g = 1$ the cortical stage leaves the retinal distortion unchanged, at $g = 2$ it fully cancels the retinal shift, and at

$g > 2$ it reverses the shift and moves the perceived hue past the undistorted color. The confusion axis is the color-space direction along which CVD observers confound hues with one another. $\delta\theta_{\text{Mach}}$ lies entirely along it, so R+C carries a single free parameter and displaces colors along one fixed direction. $\Delta\lambda$ was not fitted. Each participant was instead evaluated at three published cone-shift anchors for their subtype, 6.0, 6.5, and 8.0 nm for deutan and 1.5, 3.0, and 10.0 nm for protan. R+C results are reported at the anchor with the lowest held-out loss.

Fits (Table S13). The retinal-plus-cortical model reached the gain boundary in both participants. Anchor choice matters for the protan participant, where boundary saturation ranged from 0% to 100% across the three $\Delta\lambda$ values, so the reported fit is the anchor that generalized best. For the deutan participant, 100% of resample solutions saturated at $g \rightarrow 3.0$, and the fit was rejected at the boundary-saturation gate as model misspecification (§3.4). For the protan participant, 41% of solutions saturated at $g = 2.95$, which passed that gate, and the held-out composite favored the 2-component fit ($\bar{L}_{\text{test}} = -1.54$ against -0.86). A gain above 2 implies cortical overshoot past the undistorted hue, which the participants' confirmed residual CVD excludes.

Relation to ΔRDM . Fitting a retinal-family model directly against ΔRDM reproduced the same boundary behavior in both participants. The forward-encoding LOCO family entered neither adopted combination.

Table S13: Distortion-model fits for both CVD participants, all scored on the same held-out composite test-loss ($\bar{L}_{\text{test}}$, lower is better; §2.12; the neural RDM atom in each adopted fit is V2 for the deutan and V1 for the protan participant). The 2-component model is shown at its adopted fit. The retinal-plus-cortical (R+C) fits are reported for comparison: the deutan R+C was rejected at the boundary-saturation gate (§3.4), and the protan R+C generalized worse than the 2-component fit.

Participant	Model	Fitted parameters	Held-out fit
Deutan	2-component	$\beta_s = 6^\circ$, $\beta_c = -42^\circ$	$\bar{L}_{\text{test}} = -2.36$
Protan	2-component	$\beta_s = 2^\circ$, $\beta_c = +24^\circ$	$\bar{L}_{\text{test}} = -1.54$
Deutan	R+C	$g \rightarrow 3.0$ (100% saturated)	rejected (Gate 2)
Protan	R+C	$g = 2.95$ (41% saturated)	$\bar{L}_{\text{test}} = -0.86$

Inverse-mapping behavior. The angular form of the 2-component model is bijective by construction, so a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The filter reported in Section 3.6 is that pre-image evaluated at the eight stimulus hues.

Why the cortical gain cannot recover the distortion. The cortical gain g rescales the retinal shift uniformly, so for every value of g the R+C family displaces hues along the fixed confusion axis alone. It cannot generate the 2-component model's S-cone-axis term $\beta_s \cos(\theta - 90^\circ)$, a direction that lies 60° from the deutan confusion axis and 74° from the protan one. The observed interpolation deficit concentrates at the S-cone intermediate hues (Section 3.1), a displacement pattern this family cannot represent. This single-axis degree-of-freedom deficit accounts for the boundary behavior (§3.4). The restriction is a property of the model class and holds across fitting criteria.

S14. Identifiability checks

Four pre-specified checks bound what the fit can estimate. Test 1 assesses voxel-level parameter recovery, Test 2a measures the algorithm's noise floor, and Tests 2b and 2c assess specificity to CVD data. Each ran on the two production candidates (S08-robust, S09-primary) with PCA-basis voxel synthesis (spatial PCA(20) on SRM-projected amplitudes) as the feature space. Benjamini–Hochberg correction across the six test-bearing checks (2 candidates $\times$ 3 tests, $\alpha = 0.05$) returned no result below threshold. The production argmin is therefore treated throughout as a descriptive embedding of the CVD-specific distortion direction rather than as a point estimate with physiological magnitude.

Unselected loss atom: hV4 LOCO voxel prediction (L_{LOCO}). This atom scores how well a candidate distortion accounts for the CVD participant's own held-out responses at hV4. For each held-out color c , ridge weights W_c were estimated from the remaining seven colors across all six runs, with the penalty chosen by generalized cross-validation as in §2.7. The candidate distortion relabels each stimulus by its perceived hue, so the held-out response is predicted as $\hat{Y}_c = \mathbf{c}(\theta_c + \delta\theta_c)^\top W_c$. The fit for that color is the Pearson correlation ρ_c between $\hat{Y}_c$ and the participant's run-averaged pattern at color c . The loss averages the complement over the eight colors:

$$L_{\text{LOCO}} = \frac{1}{8} \sum_{c=1}^8 (1 - \rho_c) \quad (7)$$

Fitting the weights within participant keeps the prediction in that participant's voxel space, since the number of voxels differs across participants and HC-trained weights do not transfer.

Test 1 — Voxel-level parameter recovery. Synthetic fMRI data were generated at the production argmin (PCA-basis; 7 HC donors $\times$ 20 noise draws = 140 samples per candidate), with the psychophysical

thresholds scaled to match the distortion magnitude at the ground-truth (β_s, β_c) . The same fitting pipeline was re-run on each sample. A candidate passed when the fraction recovered within 10° on both axes (f_{10°) reached 0.5 and $|\text{bias}|$ stayed below 10° on both.

Candidate	bias (β_s, β_c)	f_{10°	Verdict
S08-robust $(6^\circ, -42^\circ)$	$(+16^\circ, -4.7^\circ)$	0.26	FAIL
S09-primary $(2^\circ, +24^\circ)$	$(+11^\circ, -27^\circ)$	0.14	FAIL

Table S14: Test 1 results. Recovery reached $f_{10^\circ} = 0.26$ and 0.14 against a pre-specified criterion of 0.5. The non-dominant axes ($|\hat{\beta}_s| \leq 6^\circ$ for both participants) fall below the effective parameter uncertainty established by Test 2a and are not recoverable. For S08-robust, the dominant axis ($|\hat{\beta}_c| = 42^\circ$, bias = 4.7°) shows lower bias than the non-dominant axis (16°), consistent with partial magnitude recoverability where the axis magnitude exceeds the noise floor.

Test 2a — Origin-recovery null. Synthetic data were generated at a ground truth of $(0, 0)$ with real HC thresholds and no synthetic psychophysical signal (PCA-basis). Real HC thresholds carry no CVD-consistent signal at the null, so this measures the noise floor free of synthesis-design contamination. A candidate passed when $|\text{bias}|$ stayed below 5° on both axes.

Candidate	$ \hat{\beta}_s _{\text{med}}$ (IQR)	$ \hat{\beta}_c _{\text{med}}$ (IQR)	β_c p95	$f_{10^\circ}^{\text{origin}}$
S08-robust	22° (40)	26° (10.5)	44°	0.00
S09-primary	16° (17.5)	24° (9)	48°	0.00

Table S15: Test 2a results. $f_{10^\circ}^{\text{origin}} = 0/140$ for both candidates: no sample was recovered within 10° of the origin under a null ground truth. Effective parameter uncertainty (PCA-basis): $\sim 20^\circ$ (β_s) and $\sim 25^\circ$ (β_c).

Test 2b — HC pseudo-CVD specificity. Each HC participant's real amplitudes were substituted as a fake CVD participant (7 carriers per candidate, deterministic), and the production CVD fit's distance from the origin was ranked among the seven HC pseudo-CVD distances ($\text{rank}_{\text{dist}}$, one-sided high). A candidate passed at rank 1.0, meaning the CVD fit exceeds every HC distance. This test is descriptive and entered no selection decision.

Candidate	$\text{rank}_{\text{dist}}$	Verdict
S08-robust	0.875	FAIL
S09-primary	0.875	FAIL

Table S16: Test 2b results. Neither candidate exceeds all HC pseudo-CVD distances.

Test 2c — Color-label permutation. CVD trial labels were shuffled over $N = 1,000$ permutations with the HC data held fixed. A candidate passed at $p_{\text{perm}} < 0.05$, one-tailed and lower.

Candidate	Real loss	Perm 5% cut	p_{perm}
S08-robust	-2.892	-3.136	0.167
S09-primary	-1.681	-3.053	0.471

Table S17: Test 2c results. Neither candidate reaches significance against the label-permutation null.

Basis caveat. All four tests use the PCA-basis loss. For the protan participant the sign of $\hat{\beta}_c$, which carries the mechanism class, holds under that loss alone. Under the SRM-basis loss the modal argmin is $(32^\circ, 0^\circ)$ in 171 of 300 resamples, leaving $\hat{\beta}_c$ positive in 17% and negative in 26%, so the two reductions return different mechanism classes for this participant. The deutan sign holds under both bases, at 300 of 300 resamples in each.

HC leave-one-out magnitude anchor. Both CVD estimates fall inside the corresponding HC range, so parameter magnitude alone is a descriptive anchor rather than a test of CVD-versus-HC specificity. Each of the $n = 7$ HC participants was refit as a pseudo-CVD case with the remaining HC pool as reference, under that participant's own production loss. The deutan combination is $\gamma_{\text{OY}} + L_{\text{RDM}}^{(\text{V2})}$ and the protan combination $\gamma_{\text{all}} + L_{\text{RDM}}^{(\text{V1})}$. The deutan estimate is $\|\hat{\beta}\| = 42.4^\circ$ against an HC range of 30.5° to 58.1° (mean 49.1°). The protan estimate is 24.1° against 23.4° to 55.5° (mean 35.7°), close to the lower bound.

Sign stability across the two preprocessing pipelines. Each participant's selected loss combination was refit on the head-motion-corrected pipeline (§S1) with every other element of the procedure held fixed. The sign of $\hat{\beta}_c$ held for the deutan participant, whose median moved from -42° to -46° with the fraction of negative resamples at 1.000 and 0.947. It did not hold for the protan participant, whose median moved from $+24^\circ$ to -12° . The primary pipeline returned no negative resample, whereas the head-motion-corrected pipeline returned 79.3% negative and a further 10.7% at the positive grid boundary. The deutan fit was also more poorly conditioned on the second pipeline, the combined fraction of resamples reaching a grid boundary rising from 0.09 to 0.72. Those boundary solutions lie on the same side as the median, so they bear on the magnitude rather than on the sign. The psychophysical atoms are unaffected by preprocessing, so the neural atom is the only component that differs between the two fits. We therefore report the deployed protan parameter as the frozen value it is and give it no physiological interpretation.

Resample structure and recovery of the production fits. Each participant's selected fit is the global minimum of its loss surface on the full data (Fig. S1). The $(32^\circ, 0^\circ)$ solution is the secondary

basin of the protan PCA landscape, so the two reductions disagree about which basin is the minimum. Refitting synthetic data with known parameters left an effective uncertainty of 22° on β_s and 26° on β_c for the deutan candidate, and 16° and 24° for the protan candidate. The deutan $|\hat{\beta}_c| = 42^\circ$ exceeds its uncertainty and was recovered to 4.7° on average. The protan $|\hat{\beta}_c| = 24^\circ$ equals its uncertainty and was recovered to 26.4° . The fitted $|\hat{\beta}_s|$ of 6° and 2° falls well below the uncertainty on that axis, so the S-cone amplitude cannot be distinguished from zero at this resolution. Recovery within 10° succeeded on 26% of draws for the deutan candidate and 14% for the protan. Within the deutan fit the two axes are constrained to different degrees. Across the 300 resamples $\hat{\beta}_c$ concentrates at -42° and -44° (202 of 300) and stays within $[-48^\circ, -36^\circ]$, whereas $\hat{\beta}_s$ distributes almost uniformly from 0° to 14° .

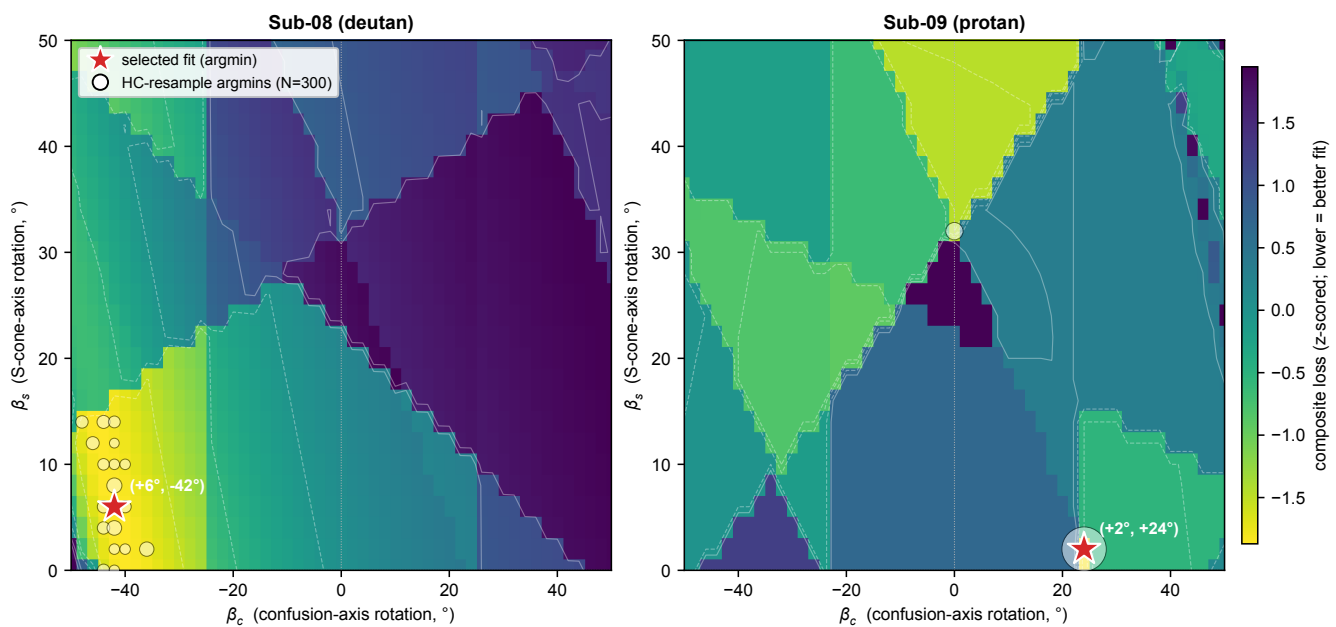


Figure S1: Per-subject loss landscape and parameter uncertainty. The z -scored composite loss over the (β_s, β_c) grid, reconstructed on the full 7-HC pool, with lower values in yellow. The loss combination is $\gamma_{OY} + L_{RDM}^{(V2)}$ for the deutan participant and $\gamma_{all} + L_{RDM}^{(V1)}$ for the protan participant. The red star marks the argmin of that surface, at $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$ for the deutan participant and $(2^\circ, +24^\circ)$ for the protan participant. It is a descriptive low-dimensional embedding of the distortion rather than a physiological point estimate. White points give the per-resample argmins over $N = 300$ HC 5-train/2-test resamples.

S15. Filter-evaluation session: design and comparator

Comparator. The deployed comparator was the macOS accessibility Color Filter (System Settings > Accessibility > Display, build 26.5.1), set at an intensity the participant self-tuned before scanning. It is the unmodified shipping product, an operating-system-level post-display transform. We compared against

it rather than a re-implemented retinal-simulation transform so that the comparator is the correction an end user actually encounters. The individualized filter was rendered directly in PsychoPy from the frozen per-subject pre-image.

Acquisition. Eight fMRI runs were acquired in ABBA-counterbalanced order, four per filter. Scanner time set this count, since six runs per condition would run roughly 90 minutes, beyond the 60–75-minute window over which participant attention and head motion stay acceptable. Four runs per condition takes about 60 minutes and sits at the recommended boundary.

Run-count adequacy. Four runs retain the Session-1 landmark, the separation between the healthy-control mean and the two CVD cases at hV4. Across all $\binom{6}{n}$ subsets of the six-run Session-1 data, hV4 LOCO adjacent accuracy held that separation at every run count down to four (Figure S2). At $n = 4$ the healthy-control mean reached 0.45 against 0.46 at six runs and stayed above the 0.25 chance level, while both CVD participants stayed below it (deutan 0.23, protan 0.14, single-case $d_{cc} < -2$ in both). A separate subsampling analysis of signal-quality and geometric-stability metrics returned the same ordering, with five runs preferred and four acceptable. Four runs nonetheless sits below the count recommended for fine-grained RDM-based model comparison. The second session therefore targets movement toward or away from the HC reference rather than adjudication between model classes.

The filter conditions each have four runs, whereas the Session-1 HC reference and the unfiltered baseline each have six. Noise in either pattern inflates disparity and RDM similarity, so an unmatched comparison penalizes the filter conditions twice, once against the baseline and once against the HC distribution that the single-case test uses. Every neural index is therefore reported run-matched. For the interpolation indices the HC reference is subsampled to four runs. For the geometry indices both the HC means and the unfiltered baseline are rebuilt from four runs, exhaustively over all $\binom{6}{4} = 15$ subsets, with the shared response model refitted within each subset and the resulting metrics averaged (`exp2_runmatched_geometry.py`). The filter conditions are never subsampled, since they only ever had four runs. The HC reference comprises $n = 7$ controls at V1–V3 and $n = 6$ at hV4, where one control has too few voxels. Matching lowers the HC disparity reference from 0.49 to 0.44 at V2 and from 0.45 to 0.43 at V1. It also raises the protan V1 RDM similarity of the unfiltered baseline from 0.25 to 0.33, which moves the individualized filter from above that baseline to below it. The ordering of conditions is otherwise unchanged on all four indices.

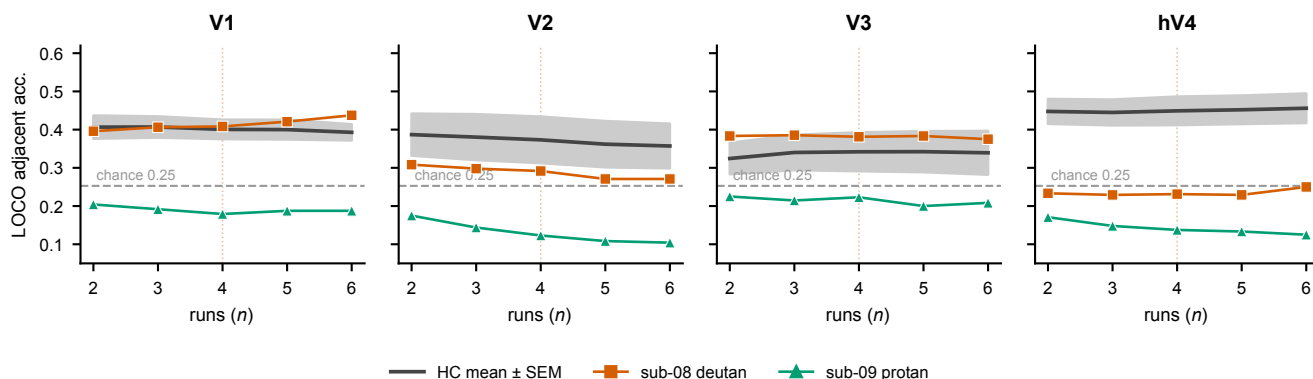


Figure S2: **Hue-interpolation adjacent accuracy as a function of the number of runs.** LOCO adjacent accuracy (canonical FE-6 basis, OLS pseudoinverse decoder) as a function of the number of runs n , computed over all $\binom{6}{n}$ run subsets of the Session-1 data, per ROI. Black line, healthy-control mean $\pm$ SEM (shaded band); dashed line, the 0.25 chance level; markers, the two CVD single cases (sub-08 deutan, sub-09 protan). At hV4 the healthy-control mean stays above chance and the CVD deficit persists at every run count, including the deployed $n = 4$ (orange guide); the metric does not depend on having the full six runs.

Single-case inference. Each condition's neural index was compared to the HC distribution as a standardized single-case effect size (Cohen's d against the HC mean and SD), reported in place of an inferential p because the grand-mean permutation null is biased for this design. Robustness was assessed by leave-one-run separation, the contrast holding after any single run is dropped, and by re-extracting both conditions on the no-filter baseline voxel set.

Psychophysical battery. Hue-discrimination thresholds were estimated with JND staircases for eight hue pairs, and the eight-alternative forced-choice (8AFC) color-identification task comprised 64 trials. Both were compared to the unfiltered Session-1 baseline and to the HC distribution.

Color classification (LORO). Table S18 reports LORO eight-way classification accuracy for every ROI and condition in both participants. Every cell stayed above the 0.125 chance level, so the displayed colors remained decodable and neither filter degraded the second-session color signal.

Forward-tuning encoding index. Figure S3 shows the LOCO forward-tuning correlation ρ to the canonical hue map, a secondary encoding index reported as corroboration only (it lies outside the validated Session-1 decoding pipeline). In the deutan participant the individualized filter reaches the HC level at hV4 ($\rho = +0.18$ against HC $+0.21$) while the deployed filter drives it negative ($\rho = -0.39$). In the

Table S18: **Second-session color classification (LORO eight-way classification accuracy).** Chance = 0.125. NF, no-filter (Session-1 baseline); Dep, deployed accessibility filter; Ind, individualized filter. HC, healthy-control mean ($n = 7$).

ROI	HC	Deutan (sub-08)			Protan (sub-09)		
		NF	Dep	Ind	NF	Dep	Ind
V1	0.71	0.79	0.84	0.72	0.79	0.91	0.66
V2	0.71	0.79	0.69	0.62	0.83	0.75	0.72
V3	0.77	0.67	0.78	0.69	0.81	1.00	0.69
hV4	0.75	0.73	0.88	0.50	0.71	0.84	0.69

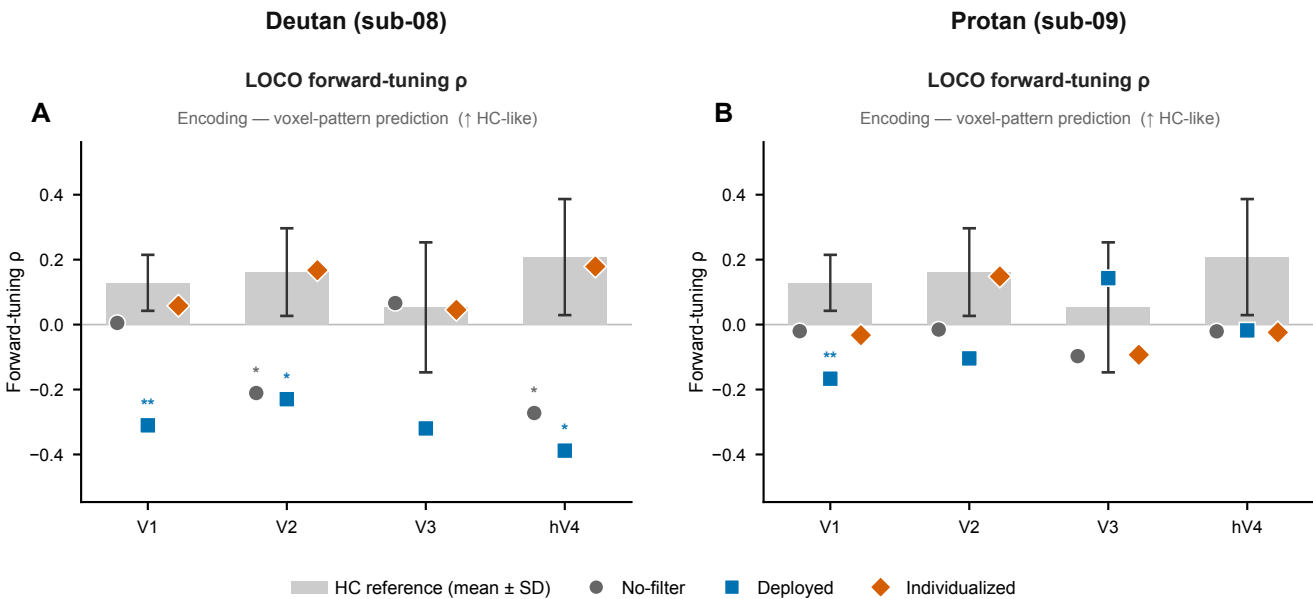


Figure S3: **Forward-tuning encoding index (corroboration only).** LOCO forward-tuning correlation ρ to the canonical hue map across V1–hV4, for the deutan (sub-08, left) and protan (sub-09, right) participants ($\uparrow$ HC-like). Gray bars, HC reference (mean $\pm$ SD, $n = 7$); markers, no-filter (gray circle), deployed accessibility filter (blue square), individualized filter (orange diamond). Stars, Crawford–Howell single-case deviation from the HC distribution (* $p < .05$, ** $p < .01$).

protan participant the three conditions do not separate, all reaching $\rho \approx -0.02$ at hV4.

S16. Statistical Analysis

Unless noted otherwise, the significance threshold was $\alpha = 0.05$. False Discovery Rate correction (Benjamini & Hochberg, 1995) was applied within two pre-specified families alone, the RDM color-pair comparisons ($q = 0.05$, 28 pairs per participant) and the six parameter-identifiability checks (section 2.13).

The exploratory per-hue vulnerability comparisons and the per-pair identification tests, including the blue-hue deficit, are reported one-tailed and uncorrected. Individual CVD participants were compared to the HC distribution using the Crawford and Howell (1998) modified t -test,

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (8)$$

with one-tailed p -values testing the directional hypothesis that CVD disparity exceeds HC. Group-level effect sizes were quantified as Hedges' g (Hedges & Olkin, 1985). Single-case effect sizes are the case-control index d_{cc} defined in §S17. Group-level permutation tests (10,000 iterations) permuted subject labels for the HC–CVD comparisons (Nichols & Holmes, 2002). Color-level permutation tests (1,000 iterations) permuted color labels within subjects for the LOCO and LORO validation.

S17. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford and Howell, 1998, whose case-control effect size is $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$, where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Effect sizes for the single-case interpolation comparisons are collected in table S19, and the geometric-disparity effect sizes are listed alongside their t statistics in table S6 (case-control $d_{cc} = t\sqrt{8/7}$, $n = 7$). Single-case comparisons of within-subject LORO accuracy against the HC distribution reached $|d_{cc}|$ between 0.25 and 1.58 across the eight participant-by-ROI tests, with all $p \geq 0.189$ (two-tailed). The largest deviation was the deutan participant at V3 ($d_{cc} = -1.58$), and at hV4 both participants gave $d_{cc} = -1.08$ ($p = 0.352$).

Cross-subject transfer of the forward encoding model was also compared between destinations. A control-trained model applied to a held-out control reached a mean accuracy of 0.526 over 28 subject-by-ROI cells. The same model applied to a CVD participant reached 0.432 over 8 cells (Mann–Whitney $U = 163.5$, $p = 0.052$, rank-biserial $r = 0.46$). Both destinations lie far above the 0.125 chance level, and the lowest single cell reached 0.271. This comparison spans participants and is therefore computed in the SRM-aligned space, unlike the within-subject readouts of §S8. sub-10 is excluded here as everywhere else.

Table S19: Effect sizes for the single-case hue-interpolation comparisons at hV4 (main text, §3.1). Crawford–Howell case-control effect size $d_{cc} = t\sqrt{(n+1)/n}$ against the $n = 7$ HC distribution, one-tailed. Per-hue tests are uncorrected and exploratory across eight hues. Both CVD participants share identical statistics at the three hues where adjacent accuracy is zero. Blue lies just above the conventional threshold and is the largest per-hue deviation.

Comparison (hV4)	t	p	d_{cc}
LOCO adjacent accuracy, deutan	−1.89	0.054	−2.02
LOCO adjacent accuracy, protan	−3.04	0.012	−3.25
<i>Per-hue adjacent accuracy (one-tailed, exploratory; both participants)</i>			
Blue	−1.93	0.051	−2.06
Purple	−0.79	0.229	−0.85
Magenta	−1.47	0.096	−1.57